Templates

Group 2 — Experience Design

- Design & UX/UI (DES)
- Interaction Design & Motion (IXD)
- Content Design & UX Writing (CDX)
- Design System & UI Tokens (DSYS)
- Information Architecture & Navigation (IAN)
- Accessibility Design (A11YD)
- Responsive Layout & Breakpoints (RLB)
- Visual Asset Production (VAP)

Experience Design — Template Set (Locked v1)

Design & UX/UI (DES)
DES-01 UX Flows (happy paths + edge cases)
DES-02 Screen Inventory (by route/screen_id)
DES-03 Screen Spec (per screen: purpose, states, actions)
DES-04 UI Component Inventory (components by ID)
DES-05 UI State Model (loading/empty/error/success)
DES-06 Form & Validation UX Rules (field rules, feedback)
DES-07 Error Handling UX (toasts, banners, inline errors, retries)
DES-08 Acceptance Hooks (screen/flow → PRD-09 criteria mapping)

Interaction Design & Motion (IXD)
IXD-01 Interaction Patterns Catalog (modals, drawers, menus, gestures)
IXD-02 Motion Rules (when/why/constraints)
IXD-03 Transition Map (screen transitions + durations)
IXD-04 Micro-interactions Spec (hover/press/drag/feedback)
IXD-05 Accessibility-Safe Motion Rules (reduce motion compliance)

Content Design & UX Writing (CDX)
CDX-01 Content Style Guide (tone, voice, terminology)
CDX-02 UI Copy Inventory (labels, buttons, helper text)
CDX-03 Empty States & Onboarding Copy
CDX-04 Error/Warning/Success Message Catalog
CDX-05 Notification Copy Templates (push/email/in-app)

Design System & UI Tokens (DSYS)
DSYS-01 Token Spec (color, type, spacing, radius, elevation)
DSYS-02 Component Variants Spec (props, variants, states)
DSYS-03 Layout Grid & Spacing Rules
DSYS-04 Iconography & Illustration Rules
DSYS-05 Theming Rules (light/dark, brand constraints)

Information Architecture & Navigation (IAN)
- IAN-01 Navigation Map (primary/secondary, tabs/drawers)
- IAN-02 Route & Deep Link Spec (route IDs, params)
- IAN-03 Information Architecture Tree (sections/pages hierarchy)
- IAN-04 Search/Filter/Sort UX (if applicable)
- IAN-05 Access-Gated Navigation Rules (role-based visibility)

Accessibility Design (A11YD)
- A11YD-01 Accessibility Requirements Checklist (WCAG-aligned)
- A11YD-02 Keyboard/Focus Order Spec
- A11YD-03 Screen Reader & Labels Spec (aria/alt/roles)
- A11YD-04 Color Contrast & Visual Accessibility Spec
- A11YD-05 Accessible Error Messaging Rules

Responsive Layout & Breakpoints (RLB)
- RLB-01 Breakpoint Definitions (sizes + names)
- RLB-02 Layout Adaptation Rules (per breakpoint)
- RLB-03 Responsive Component Behavior (tables/cards/nav)
- RLB-04 Touch Target & Density Rules
- RLB-05 Responsive Media Rules (images/video scaling)

Visual Asset Production (VAP)
- VAP-01 Asset Inventory (logos, icons, illustrations)
- VAP-02 Export Spec (formats, sizes, naming, density)
- VAP-03 Brand Usage Rules (do/don’t, clearspace)
- VAP-04 Asset Delivery Checklist (handoff requirements)
- VAP-05 Accessibility for Visual Assets (alt text, meaning)

Experience Design (Definition)

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What it is

The Experience Design group defines the **user-facing behavior** of the product: how users move through it, what they see, what they can do at each step, and the rules that govern interaction. It turns Product Definition (Group 1) into **design-authoritative specs** that downstream build templates can implement and test.

What it produces

- Screen and flow specifications that are **implementable** (states, inputs, outputs, errors)
- Navigation and information structure (routes, hierarchy, labels)
- Interaction rules (forms, validation, feedback, motion behavior)
- Design system guidance (components/tokens) that ensures UI consistency
- Content/copy requirements (microcopy, empty states, error messages)
- Accessibility requirements that are testable
- Responsive behavior rules across breakpoints
- Asset requirements and export specs for implementation

What it is used for (exactly)

- Converting **features** → **flows** → **screens** → **components** (feeds MAP + FE/MOB)
- Providing deterministic UI behavior to implement (feeds FE/MOB, QA)
- Defining navigation + IA so routing is stable (feeds ROUTE, FE-01, MOB-01)
- Defining reusable UI patterns and tokens (feeds FE-06, DSYS)
- Producing acceptance-ready UX states (feeds PRD-09, QA-02, CER)

What it is not

- Not code, repo structure, or implementation tasks (that's Group 5 / IMP)

- Not backend contracts, data schemas, or endpoint design (that's Groups 3–4)
- Not a separate source of truth for requirements; it **references** PRD/URD and does not redefine them

Scope rules

- Must reference canonical inputs: **PRD-04 feature IDs**, **PRD-09 acceptance criteria**, **URD insights**, **DMG terms**
- Must specify behavior in a way that can be tested: states, validations, error handling, accessibility rules
- Must avoid implementation detail (no file paths, no specific framework code), but **must** be precise enough that implementation is straightforward

Subcategory meaning (one-line each)

- **DES**: Core flows/screens/states/component inventory specs
- **IXD**: Interaction mechanics + motion rules (what animates, when, why)
- **CDX**: UX writing + content requirements (copy, tone, system messages)
- **DSYS**: Tokens/components/variants rules (design system contract)
- **IAN**: App structure + navigation model (hierarchy, route naming, labels)
- **A11YD**: Accessibility requirements (WCAG-aligned, testable checks)
- **RLB**: Responsive rules (breakpoints, layout adaptations)
- **VAP**: Asset production requirements (formats, sizes, naming, export specs)

Design & UX/UI (DES)

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- DES-01 UX Flows (happy paths + edge cases)
- DES-02 Screen Inventory (by route/screen_id)
- DES-03 Screen Spec (per screen: purpose, states, actions)
- DES-04 UI Component Inventory (components by ID)
- DES-05 UI State Model (loading/empty/error/success)
- DES-06 Form & Validation UX Rules (field rules, feedback)
- DES-07 Error Handling UX (toasts, banners, inline errors, retries)
- DES-08 Acceptance Hooks (screen/flow → PRD-09 criteria mapping)

DES-01

DES-01 — UX Flows (happy paths + edge cases)

Header Block

- template_id: DES-01
- title: UX Flows (happy paths + edge cases)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-01_UX_Flows.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["PRD-04", "PRD-05", "PRD-09"]
- inputs_required: ["PRD-04", "PRD-05", "PRD-09", "DMG-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the user-facing workflows as step-by-step flows that can be implemented and tested. Each flow must map to feature IDs and acceptance criteria, include edge cases, and remain UI-framework agnostic while being behavior-specific.

Inputs Required

- PRD-04: {{xref:PRD-04}}
- PRD-05: {{xref:PRD-05}} | OPTIONAL
- PRD-09: {{xref:PRD-09}} | OPTIONAL
- DMG-01: {{xref:DMG-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Flow list (minimum 5 for non-trivial products)
- For each flow:
    - flow_id
    - name
    - primary persona(s)
    - linked_feature_ids
    - preconditions

    - happy-path steps (ordered)
    - edge cases (minimum 2)
    - entry points (where flow starts)
    - exit outcomes (success/fail)
    - system feedback (loading/success/error states)
    - linked acceptance criteria IDs (ac_*)
- Flow coverage: every P0 feature has at least one flow mapping

Optional Fields

- Variants (mobile vs web differences) | OPTIONAL
- Timing/latency expectations (qualitative) | OPTIONAL
- Notes | OPTIONAL

Rules

- Flows must use existing IDs: feature IDs from PRD-04; acceptance IDs from PRD-09.
- Steps must be written as user actions + system response (both sides).
- Do not define UI layout; reference screen IDs when known (DES-02/03).
- Edge cases must include at least:
    - validation failure or missing input
    - permission/entitlement failure OR system failure (network/server)
- If a flow touches sensitive data, reference relevant privacy/security constraints (DGP/SEC) via pointers only.

Output Format

1) Flow Index (summary)

flow_id	name	persona_ids	feature_ids	entry_point	exit_outcomes	mapped_ac_ids
flow_01	{{flows[0].name}}	{{flows[0].persona_ids}}	{{flows[0].feature_ids}}	{{flows[0].entry}}	{{flows[0].exits}}	{{flows[0].ac_ids}}
flow_02	{{flows[1].name}}	{{flows[1].persona_ids}}	{{flows[1].feature_ids}}	{{flows[1].entry}}	{{flows[1].exits}}	{{flows[1].ac_ids}}

2) Flow Detail Blocks (required, one per flow)

{{flows[0].flow_id}} — {{flows[0].name}}

- **Primary personas:** {{flows[0].persona_ids}}
- **Linked features:** {{flows[0].feature_ids}}
- **Preconditions:** {{flows[0].preconditions}}
- **Entry points:** {{flows[0].entry_points}}
- **Exit outcomes:** {{flows[0].exit_outcomes}}
- **Related screens:** {{flows[0].screen_ids}} | OPTIONAL
- **Related endpoints:** {{flows[0].endpoint_ids}} | OPTIONAL

Happy Path (steps)

1. User: {{flows[0].happy_steps[0].user_action}}
   System: {{flows[0].happy_steps[0].system_response}}
2. User: {{flows[0].happy_steps[1].user_action}}
   System: {{flows[0].happy_steps[1].system_response}}

Edge Cases (minimum 2)

- ec_01: {{flows[0].edge_cases[0].condition}}
  - User sees: {{flows[0].edge_cases[0].ux_feedback}}
  - System does: {{flows[0].edge_cases[0].system_behavior}}
- ec_02: {{flows[0].edge_cases[1].condition}}
  - User sees: {{flows[0].edge_cases[1].ux_feedback}}
  - System does: {{flows[0].edge_cases[1].system_behavior}}

System Feedback / States

- Loading: {{flows[0].states.loading}}
- Success: {{flows[0].states.success}}
- Error: {{flows[0].states.error}}

Acceptance Criteria Mapping

- {{flows[0].ac_ids}}

Variants (optional)

- {{flows[0].variants}} | OPTIONAL

3) P0 Feature Coverage (required)

feature_id	feature_name	has_flow	mapped_flow_ids
{{spec.features_by_id[feat_p0].id}}	{{spec.features_by_id[feat_p0].name}}	{{coverage[feat_p0].covered}}	{{coverage[feat_p0].flow_ids}}

4) Notes (optional)

- {{notes[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-04}}, {{xref:PRD-05}} | OPTIONAL, {{xref:PRD-09}} | OPTIONAL
- Downstream: {{xref:DES-02}}, {{xref:DES-03}}, {{xref:MAP-01}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Minimum 5 flows; simple steps + 2 edge cases each.
- intermediate: Required. Add acceptance criteria mapping and entry/exit outcomes.
- advanced: Required. Add related screens/endpoints and variants where needed.

Unknown Handling

- UNKNOWN_ALLOWED: `screen_ids`, `endpoint_ids`, `variants`, `timing_expectations`, `notes`
- If any P0 feature has no flow mapping → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
  - required_fields_present == true
  - flows_count >= 5
  - every_flow_has_edge_cases == true
  - every_p0_feature_has_flow == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DES-02

DES-02 — Screen Inventory (by route/screen_id)

Header Block

- template_id: DES-02
- title: Screen Inventory (by route/screen_id)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-02_Screen_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.SCREEN
- upstream_dependencies: ["DES-01", "PRD-04"]
- inputs_required: ["DES-01", "PRD-04", "IAN-01", "IAN-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical inventory of screens, their IDs, and how they relate to routes/navigation. This is the bridge between flows and implementation routing.

Inputs Required

- DES-01: {{xref:DES-01}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- IAN-01: {{xref:IAN-01}} | OPTIONAL
- IAN-02: {{xref:IAN-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Screen list (minimum 8 for non-trivial products)
- For each screen:
    - screen_id
    - name
    - route_id / path (or nav key)
    - purpose (1–2 sentences)
    - entry points (what routes/actions lead here)
    - exit destinations (where user can go next)

    - linked_flow_ids
    - linked_feature_ids
    - access requirements (role/entitlement)
    - primary states (from DES-05)
- Coverage: every flow references at least one screen OR explicitly marks screenless (background flow)

Optional Fields

- Platform variants (web/mobile) | OPTIONAL
- Notes | OPTIONAL

Rules

- Screen IDs must be stable and unique (`screen_<slug>`).
- Route IDs must match IAN-02 if present; otherwise mark UNKNOWN and link to IAN work.
- Access requirements must reference PRD-03/IAM concepts, not invent new roles.
- Do not define component-level layout; that belongs in DES-03/04.

Output Format

1) Screen Inventory (canonical)

screen_id	**name**	**route_id_or_path**	**purpose**	**entry_points**	**exit_destinations**	**flow_ids**	**feature_ids**	**access_req**	**primary_states**	**platform**	**notes**
screen_01	{{screens[0].name}}	{{screens[0].route}}	{{screens[0].purpose}}	{{screens[0].entry}}	{{screens[0].exit}}	{{screens[0].flow_ids}}	{{screens[0].feature_ids}}	{{screens[0].access}}	{{screens[0].states}}	{{screens[0].platform}}	{{screens[0].notes}}

screen_02	{{screens[1].name}}	{{screens[1].route}}	{{screens[1].purpose}}	{{screens[1].entry}}	{{screens[1].exit}}	{{screens[1].flow_ids}}	{{screens[1].feature_ids}}	{{screens[1].access}}	{{screens[1].states}}	{{screens[1].platform}}	{{screens[1].notes}}

2) Flow → Screen Coverage (required)

flow_id	flow_name	mapped_screen_ids	notes
{{flows[0].flow_id}}	{{flows[0].name}}	{{flows[0].screen_ids}}	{{flows[0].notes}}

Cross-References

- Upstream: {{xref:DES-01}}, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:DES-03}}, {{xref:IAN-02}} | OPTIONAL, {{xref:MAP-01}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:MOB-01}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Inventory + routes + flow mapping.
- intermediate: Required. Add access requirements and primary states.
- advanced: Required. Add platform variants and cleaner entry/exit mapping.

Unknown Handling

- UNKNOWN_ALLOWED: `route_id_or_path` (if IAN-02 not ready), `platform`, `notes`
- If any screen_id is duplicated or missing → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SCREEN
- Pass conditions:
  - required_fields_present == true
  - screens_count >= 8
  - screen_ids_unique == true

- flow_screen_coverage_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DES-03

DES-03 — Screen Spec (per screen: purpose, states, actions)

Header Block

- template_id: DES-03
- title: Screen Spec (per screen: purpose, states, actions)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-03_Screen_Specs.md
- compliance_gate_id: TMP-05.PRIMARY.SCREEN
- upstream_dependencies: ["DES-02", "DES-05", "PRD-09"]
- inputs_required: ["DES-02", "DES-05", "PRD-09", "DMG-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define implementable behavior for each screen: purpose, data needs, UI states, actions, validations, and error handling. This is the authoritative behavioral spec for frontend/mobile implementation.

Inputs Required

- DES-02: {{xref:DES-02}}
- DES-05: {{xref:DES-05}} | OPTIONAL
- PRD-09: {{xref:PRD-09}} | OPTIONAL
- DMG-01: {{xref:DMG-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- One spec block per screen_id
- For each screen:
    - purpose
    - entry conditions
    - data required (entities/fields)
    - primary UI states (loading/empty/error/success)
    - actions (user actions + system responses)

    - validation rules (if any)
    - error cases + user feedback
    - navigation outcomes (where actions go)
    - mapped acceptance criteria IDs
    - access requirements
- State coverage: each screen declares behavior for loading + error + empty (if applicable)

Optional Fields

- Analytics events (screen_view, action events) | OPTIONAL
- Performance expectations (qualitative) | OPTIONAL
- Notes | OPTIONAL

Rules

- No layout pixel specs; describe structure as regions (header/body/footer, list/detail, etc.).
- Actions must be deterministic: action → system response → next state.
- Acceptance criteria must reference PRD-09 IDs when available.

Output Format

Screen: {{screen_specs[0].screen_id}} — {{screen_specs[0].name}}

- **Purpose:** {{screen_specs[0].purpose}}
- **Entry conditions:** {{screen_specs[0].entry_conditions}}
- **Access requirements:** {{screen_specs[0].access}}
- **Data required:** {{screen_specs[0].data_required}}

States

- Loading: {{screen_specs[0].states.loading}}
- Empty: {{screen_specs[0].states.empty}} | OPTIONAL
- Success: {{screen_specs[0].states.success}}
- Error: {{screen_specs[0].states.error}}

Actions

1. User action: {{screen_specs[0].actions[0].user_action}}
   System response: {{screen_specs[0].actions[0].system_response}}
   Next state: {{screen_specs[0].actions[0].next_state}}
   Navigation: {{screen_specs[0].actions[0].navigation}} | OPTIONAL

Validation (if applicable)

- {{screen_specs[0].validation_rules}} | OPTIONAL

Error Cases

- {{screen_specs[0].error_cases[0]}}
- {{screen_specs[0].error_cases[1]}} | OPTIONAL

Acceptance Criteria

- {{screen_specs[0].ac_ids}}

Analytics (optional)

- {{screen_specs[0].analytics}} | OPTIONAL

Notes

- {{screen_specs[0].notes}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-02}}, {{xref:DES-05}} | OPTIONAL, {{xref:PRD-09}} | OPTIONAL
- Downstream: {{xref:FE-02}}, {{xref:FORM-*}}* | *OPTIONAL, {{xref:CER-}}* | OPTIONAL, {{xref:MAP-01}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define purpose, actions, and basic states.
- intermediate: Required. Add validation + error cases + navigation outcomes.
- advanced: Required. Add data-required mapping and analytics hooks.

Unknown Handling

- UNKNOWN_ALLOWED: `analytics`, `performance_expectations`, `notes`, `empty_state` (if not applicable)
- If a screen lacks loading+error behavior → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SCREEN
- Pass conditions:
    - required_fields_present == true

- every_screen_has_states == true
- actions_present_for_each_screen == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DES-04

DES-04 — UI Component Inventory (components by ID)

Header Block

- template_id: DES-04
- title: UI Component Inventory (components by ID)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-04_UI_Component_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["DES-03"]
- inputs_required: ["DES-03", "DSYS-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Create the canonical list of UI components (by ID) needed to implement the screens and flows. Components here are functional building blocks, not styling tokens.

Inputs Required

- DES-03: {{xref:DES-03}}
- DSYS-02: {{xref:DSYS-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Component list (minimum 10 for non-trivial products)
- For each component:
  - component_id
  - name
  - purpose
  - used_on_screen_ids
  - states (loading/disabled/error/etc.)
  - props/inputs (conceptual)
  - outputs/events (what it emits)
  - accessibility notes (if any)

    - variant notes (if any)
- Coverage: every screen references at least one component OR explicitly marks “layout-only”

Optional Fields

- Reuse candidates (existing library mapping) | OPTIONAL
- Notes | OPTIONAL

Rules

- Component IDs must be stable (`comp_<slug>`).
- Props are conceptual; implementation details live in FE templates.
- If using design system components, reference DSYS component names/variants.

Output Format

1) Component Inventory (canonical)

component_id	name	purpose	screen_ids	states	props_inputs	outputs_events	a11y_notes	variants	reuse_candidate	notes
comp_01	{{components[0].name}}	{{components[0].purpose}}	{{components[0].screen_ids}}	{{components[0].states}}	{{components[0].props}}	{{components[0].outputs}}	{{components[0].a11y}}	{{components[0].variants}}	{{components[0].reuse}}	{{components[0].notes}}

2) Screen → Component Coverage (required)

screen_id	screen_name	mapped_component_ids	notes
{{screens[0].screen_id}}	{{screens[0].name}}	{{screens[0].component_ids}}	{{screens[0].notes}}

Cross-References

- Upstream: {{xref:DES-03}}, {{xref:DSYS-02}} | OPTIONAL

- Downstream: {{xref:FE-02}}, {{xref:MAP-01}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Inventory + purpose + screens.
- intermediate: Required. Add states and inputs/outputs.
- advanced: Not required. (Advanced component engineering belongs in FE/FPLAT.)

Unknown Handling

- UNKNOWN_ALLOWED: `reuse_candidate`, `variants`, `a11y_notes`, `notes`
- If a component_id is missing or duplicated → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
  - required_fields_present == true
  - components_count >= 10
  - component_ids_unique == true
  - screen_component_coverage_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DES-05

DES-05 — UI State Model (loading/empty/error/success)

Header Block

- template_id: DES-05
- title: UI State Model (loading/empty/error/success)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-05_UI_State_Model.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["DES-01", "PRD-06"]
- inputs_required: ["DES-01", "PRD-06", "CDX-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical UX behavior for common UI states so screens are consistent: loading, empty, error, success, disabled, offline, and permission denied.

Inputs Required

- DES-01: {{xref:DES-01}}
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- CDX-04: {{xref:CDX-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- State catalog (minimum: loading, empty, error, success, offline, permission_denied)
- For each state:
    - state_id
    - definition
    - when it occurs (conditions)
    - user feedback (what user sees)
    - retry/recovery action (if any)
    - logging/telemetry expectation (optional pointer)
    - accessibility considerations

- Default precedence rules (which state wins when multiple apply)

Optional Fields

- Component-level state guidelines | OPTIONAL
- Notes | OPTIONAL

Rules

- Error states must define user action options (retry, back, support, etc.).
- Offline behavior must be consistent with MOB-03 (if mobile/offline).
- Permission denied must align to IAM/BRP entitlements.
- Avoid duplicating copy; reference CDX-04 for exact messages if available.

Output Format

1) State Catalog (canonical)

state_id	definition	when_occurs	user_feedback	recovery_action	a11y_notes	telemetry_notes
loading	{{states.loading.definition}}	{{states.loading.when}}	{{states.loading.feedback}}	{{states.loading.recovery}}	{{states.loading.a11y}}	{{states.loading.telemetry}}
empty	{{states.empty.definition}}	{{states.empty.when}}	{{states.empty.feedback}}	{{states.empty.recovery}}	{{states.empty.a11y}}	{{states.empty.telemetry}}
error	{{states.error.definition}}	{{states.error.when}}	{{states.error.feedback}}	{{states.error.recovery}}	{{states.error.a11y}}	{{states.error.telemetry}}
success	{{states.success.definition}}	{{states.success.when}}	{{states.success.feedback}}	{{states.success.recovery}}	{{states.success.a11y}}	{{states.success.telemetry}}
offline	{{states.offline.definition}}	{{states.offline.when}}	{{states.offline.feedback}}	{{states.offline.recovery}}	{{states.offline.a11y}}	{{states.offline.telemetry}}
permission_denied	{{states.permission_denied.definition}}	{{states.permission_denied.when}}	{{states.permission_denied.feedback}}	{{states.permission_denied.recovery}}	{{states.permission_denied.a11y}}	{{states.permission_denied.telemetry}}

2) Precedence Rules (required)

- {{precedence[0]}}
- {{precedence[1]}} | OPTIONAL

3) Notes (optional)

- {{notes[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-01}}, {{xref:PRD-06}} | OPTIONAL
- Downstream: {{xref:DES-03}}, {{xref:FE-07}} | OPTIONAL, {{xref:CER-*}} | OPTIONAL, {{xref:MOB-03}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define state behavior and user feedback.
- intermediate: Required. Add recovery actions and precedence rules.
- advanced: Required. Add accessibility and telemetry expectations.

Unknown Handling

- UNKNOWN_ALLOWED: `telemetry_notes`, `component_level_guidelines`, `notes`
- If error state lacks recovery_action → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
  - required_fields_present == true
  - minimum_states_present == true
  - precedence_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DES-06

DES-06 — Form & Validation UX Rules (field rules, feedback)

Header Block

- template_id: DES-06
- title: Form & Validation UX Rules (field rules, feedback)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-06_Form_Validation_UX.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["DES-03", "BRP-01"]
- inputs_required: ["DES-03", "BRP-01", "CDX-02", "CDX-04", "A11YD-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define consistent UX rules for forms and validation across the product: when validation happens, how errors are shown, how success is confirmed, and how accessibility is handled.

Inputs Required

- DES-03: {{xref:DES-03}}
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- CDX-02: {{xref:CDX-02}} | OPTIONAL
- CDX-04: {{xref:CDX-04}} | OPTIONAL
- A11YD-05: {{xref:A11YD-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Validation timing rules (on change/on blur/on submit)
- Field error presentation rules (inline, summary, toast)
- Required field marking rules
- Cross-field validation behavior
- Server-side validation error handling
- Success confirmation rules

- Disabled/readonly field behavior
- Accessibility rules (focus management, announcements)
- Copy guidance pointers (error message source)

Optional Fields

- Form autosave rules | OPTIONAL
- Draft/recovery rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Validation must be deterministic and consistent across screens.
- Errors must be actionable; no generic “invalid” without guidance.
- If server validation fails, UI must map to field(s) when possible.
- Copy should be referenced from CDX catalogs when available.

Output Format

1) Validation Timing (required)

- on_change: {{validation_timing.on_change}}
- on_blur: {{validation_timing.on_blur}}
- on_submit: {{validation_timing.on_submit}}
- mixed_rules: {{validation_timing.mixed}} | OPTIONAL

2) Error Presentation (required)

- Inline field errors: {{error_ui.inline}}
- Form-level summary: {{error_ui.summary}}
- Toast/banner usage: {{error_ui.toast_banner}}
- When to block submit: {{error_ui.block_submit}}

3) Required Fields (required)

- Indicator: {{required_fields.indicator}}
- When shown: {{required_fields.when_shown}}

4) Cross-field Validation (required)

- {{cross_field[0]}}
- {{cross_field[1]}} | OPTIONAL

5) Server-side Validation Errors (required)

- Mapping rule: {{server_validation.mapping_rule}}
- Unknown field error fallback: {{server_validation.fallback}}
- Retry guidance: {{server_validation.retry_guidance}} | OPTIONAL

6) Success Confirmation (required)

- Inline confirmation: {{success.inline}}
- Toast/banner confirmation: {{success.toast_banner}}
- Navigation after success: {{success.nav_after}} | OPTIONAL

7) Accessibility Rules (required)

- Focus on first error: {{a11y.focus_first_error}}
- Announce errors: {{a11y.announce_errors}}
- Error summary semantics: {{a11y.summary_semantics}} | OPTIONAL

8) Copy Source (required)

- Error copy: {{xref:CDX-04}} | OPTIONAL
- Label copy: {{xref:CDX-02}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-03}}, {{xref:BRP-01}} | OPTIONAL
- Downstream: {{xref:FORM-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Timing + error presentation + required fields.
- intermediate: Required. Add server validation mapping and accessibility rules.
- advanced: Required. Add autosave/draft rules and clear fallback logic.

Unknown Handling

- UNKNOWN_ALLOWED: `autosave_rules`, `draft_recovery`, `mixed_rules`, `notes`
- If error presentation rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
  - required_fields_present == true
  - validation_timing_present == true
  - error_presentation_present == true
  - accessibility_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DES-07

DES-07 — Error Handling UX (toasts, banners, inline errors, retries)

Header Block

- template_id: DES-07
- title: Error Handling UX (toasts, banners, inline errors, retries)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-07_Error_Handling_UX.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["DES-05", "ARC-06", "API-03"]
- inputs_required: ["DES-05", "ARC-06", "API-03", "CDX-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define consistent UX rules for error presentation and recovery across the app: where errors appear, what they say, whether actions are available, and how retries/backoff behave.

Inputs Required

- DES-05: {{xref:DES-05}}
- ARC-06: {{xref:ARC-06}} | OPTIONAL
- API-03: {{xref:API-03}} | OPTIONAL
- CDX-04: {{xref:CDX-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Error surface rules (inline vs banner vs toast vs modal)
- Error categories (validation/network/server/permission/unknown)
- Retry rules (when allowed, backoff, max attempts)
- Fallback rules (when to show generic message)
- Logging/telemetry pointer rules (what gets recorded)
- Accessibility rules (announcements, focus)

Optional Fields

- Support escalation UI (contact support) | OPTIONAL
- Offline-specific errors | OPTIONAL
- Notes | OPTIONAL

Rules

- Error messages must be actionable and consistent; do not leak sensitive details.
- Permission errors must align to IAM/BRP entitlements.
- Server errors should map reason codes when available (ARC-06).
- Retry must not create destructive repeats unless idempotent.

Output Format

1) Error Surface Rules (required)

error_type	preferred_surface	when_used	user_action	notes
validation	{{surfaces.validation.surface}}	{{surfaces.validation.when}}	{{surfaces.validation.action}}	{{surfaces.validation.notes}}
network	{{surfaces.network.surface}}	{{surfaces.network.when}}	{{surfaces.network.action}}	{{surfaces.network.notes}}
server	{{surfaces.server.surface}}	{{surfaces.server.when}}	{{surfaces.server.action}}	{{surfaces.server.notes}}
permission	{{surfaces.permission.surface}}	{{surfaces.permission.when}}	{{surfaces.permission.action}}	{{surfaces.permission.notes}}
unknown	{{surfaces.unknown.surface}}	{{surfaces.unknown.when}}	{{surfaces.unknown.action}}	{{surfaces.unknown.notes}}

2) Retry Rules (required)

- When retry is shown: {{retry.when}}
- Backoff: {{retry.backoff}}
- Max attempts: {{retry.max_attempts}}
- Idempotency note: {{retry.idempotency_note}} | OPTIONAL

3) Fallback Rules (required)

- Generic message conditions: {{fallback.when_generic}}
- Sensitive detail redaction: {{fallback.redaction}}
- Unknown error path: {{fallback.unknown_path}}

4) Accessibility (required)

- Announce errors: {{a11y.announce}}
- Focus management: {{a11y.focus}}
- Toast timing considerations: {{a11y.toast_timing}} | OPTIONAL

5) Logging/Telemetry (required)

- What to log: {{telemetry.what}}
- Correlation ID display policy: {{telemetry.correlation_id_policy}} | OPTIONAL

6) Support Escalation (optional)

- {{support.ui}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-05}}, {{xref:ARC-06}} | OPTIONAL, {{xref:API-03}} | OPTIONAL
- Downstream: {{xref:CER-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL, {{xref:OBS-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define surfaces and retry basics.
- intermediate: Required. Add fallback and telemetry rules.
- advanced: Required. Add idempotency constraints and support escalation patterns.

Unknown Handling

- UNKNOWN_ALLOWED: `support_escalation_ui`, `offline_specific`, `notes`
- If retry rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
    - required_fields_present == true

- surfaces_defined == true
- retry_rules_present == true
- fallback_rules_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DES-08

DES-08 — Acceptance Hooks (screen/flow → PRD-09 criteria mapping)

Header Block

- template_id: DES-08
- title: Acceptance Hooks (screen/flow → PRD-09 criteria mapping)
- type: design_ux
- template_version: 1.0.0
- output_path: 10_app/design/DES-08_Acceptance_Hooks.md
- compliance_gate_id: TMP-05.PRIMARY.DESIGN
- upstream_dependencies: ["DES-01", "DES-02", "PRD-09"]
- inputs_required: ["DES-01", "DES-02", "PRD-09", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create a deterministic mapping from acceptance criteria (PRD-09) to the UX surfaces where they are satisfied (flows/screens) so QA and implementation can prove completion without ambiguity.

Inputs Required

- DES-01: {{xref:DES-01}}
- DES-02: {{xref:DES-02}}
- PRD-09: {{xref:PRD-09}}
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Mapping table:
  - ac_id → flow_id(s) + screen_id(s)
- Coverage check:
  - every P0 acceptance criterion has at least one hook
- Notes on multi-surface criteria (where multiple screens satisfy one criterion)

Optional Fields

- Endpoint/component pointers | OPTIONAL
- Test case pointers | OPTIONAL
- Notes | OPTIONAL

Rules

- Only reference existing IDs: ac_* from PRD-09, flow_* from DES-01, screen_* from DES-02.
- This does not define tests; it defines where criteria are satisfied.
- If any P0 acceptance criterion lacks a hook, gate fails.

Output Format

1) Acceptance Hooks Map (canonical)

ac_id	feature_id	mapped_flow_ids	mapped_screen_ids	notes	mapped_endpoint_ids	mapped_component_ids
{{hooks[0].ac_id}}	{{hooks[0].feature_id}}	{{hooks[0].flow_ids}}	{{hooks[0].screen_ids}}	{{hooks[0].notes}}	{{hooks[0].endpoint_ids}}	{{hooks[0].component_ids}}

2) Coverage (required)

priority	total_ac	hooked_ac	missing_ac_ids
P0	{{coverage.p0.total}}	{{coverage.p0.hooked}}	{{coverage.p0.missing}}
P1	{{coverage.p1.total}}	{{coverage.p1.hooked}}	{{coverage.p1.missing}}

Cross-References

- Upstream: {{xref:PRD-09}}, {{xref:DES-01}}, {{xref:DES-02}}
- Downstream: {{xref:QA-02}} | OPTIONAL, {{xref:MAP-04}} | OPTIONAL, {{xref:TRC-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Map P0 criteria to at least one flow and screen.
- intermediate: Required. Add feature_id and multi-surface notes.
- advanced: Required. Add endpoint/component pointers when known.

Unknown Handling

- UNKNOWN_ALLOWED: `mapped_endpoint_ids`, `mapped_component_ids`, `notes`, `test_case_pointers`
- If any P0 ac_id is missing mapping → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DESIGN
- Pass conditions:
  - required_fields_present == true
  - every_p0_ac_has_hook == true
  - ids_reference_valid_sources == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Interaction Design & Motion (IXD)

Interaction Design & Motion (IXD)

- IXD-01 Interaction Patterns Catalog (modals, drawers, menus, gestures)
- IXD-02 Motion Rules (when/why/constraints)
- IXD-03 Transition Map (screen transitions + durations)
- IXD-04 Micro-interactions Spec (hover/press/drag/feedback)
- IXD-05 Accessibility-Safe Motion Rules (reduce motion compliance)

IXD-01

IXD-01 — Interaction Patterns Catalog (modals, drawers, menus, gestures)

Header Block

- template_id: IXD-01
- title: Interaction Patterns Catalog (modals, drawers, menus, gestures)
- type: interaction_design_motion
- template_version: 1.0.0
- output_path: 10_app/design/IXD-01_Interaction_Patterns_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.IXD
- upstream_dependencies: ["DES-01", "DES-03", "DES-06", "DES-07"]
- inputs_required: ["DES-01", "DES-03", "DES-06", "DES-07", "DSYS-02", "A11YD-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical catalog of interaction patterns used across the product so interactions are consistent, learnable, and implementable. This is the behavioral contract for UI mechanics (not visual styling), and it must include accessibility expectations and default fallbacks.

Inputs Required

- DES-01: {{xref:DES-01}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL
- DES-06: {{xref:DES-06}} | OPTIONAL
- DES-07: {{xref:DES-07}} | OPTIONAL
- DSYS-02: {{xref:DSYS-02}} | OPTIONAL
- A11YD-02: {{xref:A11YD-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Pattern list (minimum 12 for non-trivial products)
- For each pattern:
    - pattern_id
    - name

    - category (modal/drawer/menu/tooltip/toast/list/detail/wizard/gesture/etc.)
    - purpose (why it exists)
    - when_to_use
    - when_not_to_use
    - trigger (what opens/starts it)
    - dismissal/exit rules (how it closes/ends)
    - primary states (loading/empty/error/disabled)
    - accessibility requirements (focus trap, escape key, announcements, etc.)
    - platform notes (web/mobile differences)
    - mapping to components (optional: component_id)
- Global consistency rules (priorities when patterns overlap)

Optional Fields

- Anti-patterns (explicitly forbidden variants) | OPTIONAL
- Notes | OPTIONAL

Rules

- Pattern IDs must be stable and unique (`pat_<slug>`).
- Must not conflict with DES-05 state rules or DES-07 error rules; reference them.
- Any pattern that traps focus must define:
    - focus entry point
    - focus loop rules
    - escape behavior
- Any gesture pattern must define a non-gesture fallback (accessibility + desktop).

Output Format

1) Pattern Index (summary)

pattern_id	name	category	primary_use	key_accessibility_requirement	platform
pat_modal	{{patterns[0].name}}	{{patterns[0].category}}	{{patterns[0].when_to_use_short}}	{{patterns[0].a11y_key}}	{{patterns[0].platform}}
pat_drawer	{{patterns[1].name}}	{{patterns[1].category}}	{{patterns[1].when_to_use_short}}	{{patterns[1].a11y_key}}	{{patterns[1].platform}}

2) Pattern Detail Blocks (required, one per pattern)

{{patterns[0].pattern_id}} — {{patterns[0].name}}

- **Category:** {{patterns[0].category}}
- **Purpose:** {{patterns[0].purpose}}

When to use

- {{patterns[0].when_to_use[0]}}
- {{patterns[0].when_to_use[1]}} | OPTIONAL

When NOT to use

- {{patterns[0].when_not_to_use[0]}}
- {{patterns[0].when_not_to_use[1]}} | OPTIONAL

Trigger

- {{patterns[0].trigger}}

Dismissal / Exit Rules

- Primary dismissal: {{patterns[0].dismissal.primary}}
- Secondary dismissal: {{patterns[0].dismissal.secondary}} | OPTIONAL
- Destructive action confirmation rule: {{patterns[0].dismissal.destructive_confirm}} | OPTIONAL

States

- Loading: {{patterns[0].states.loading}}
- Error: {{patterns[0].states.error}}
- Empty: {{patterns[0].states.empty}} | OPTIONAL
- Disabled: {{patterns[0].states.disabled}} | OPTIONAL

Accessibility Requirements

- Focus behavior: {{patterns[0].a11y.focus_behavior}}
- Keyboard support: {{patterns[0].a11y.keyboard}}
- Screen reader announcements: {{patterns[0].a11y.announcements}}
- Reduced motion considerations: {{patterns[0].a11y.reduced_motion}} | OPTIONAL

Platform Notes

- Web: {{patterns[0].platform_notes.web}} | OPTIONAL
- Mobile: {{patterns[0].platform_notes.mobile}} | OPTIONAL
- Gesture fallback (if applicable): {{patterns[0].platform_notes.gesture_fallback}} | OPTIONAL

Component Mapping (optional)

- component_id: {{patterns[0].component_id}} | OPTIONAL

References

- State model: {{xref:DES-05}} | OPTIONAL
- Error handling: {{xref:DES-07}} | OPTIONAL

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3) Global Consistency Rules (required)

- Pattern precedence when multiple apply: {{global_rules.precedence}}
- Default pattern for confirmations: {{global_rules.confirmation_default}}
- Default pattern for non-blocking feedback: {{global_rules.nonblocking_feedback_default}}
- Default pattern for blocking errors: {{global_rules.blocking_error_default}}

4) Anti-Patterns (optional)

- {{anti_patterns[0]}} | OPTIONAL
- {{anti_patterns[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-03}} | OPTIONAL, {{xref:DES-06}} | OPTIONAL, {{xref:DES-07}} | OPTIONAL
- Downstream: {{xref:DSYS-02}} | OPTIONAL, {{xref:FE-02}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define 12 patterns with triggers + dismissal + a11y basics.
- intermediate: Required. Add when_not_to_use and platform fallbacks.
- advanced: Required. Add global precedence rules and anti-patterns.

Unknown Handling

- UNKNOWN_ALLOWED: `component_mapping`, `anti_patterns`, `platform_notes`, `notes`

- If any pattern lacks dismissal rules or accessibility requirements → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IXD
- Pass conditions:
  - required_fields_present == true
  - patterns_count >= 12
  - every_pattern_has_trigger_and_dismissal == true
  - every_pattern_has_a11y_requirements == true
  - global_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

IXD-02

IXD-02 — Motion Rules (when/why/constraints)

Header Block

- template_id: IXD-02
- title: Motion Rules (when/why/constraints)
- type: interaction_design_motion
- template_version: 1.0.0
- output_path: 10_app/design/IXD-02_Motion_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.IXD
- upstream_dependencies: ["IXD-01", "DES-05", "DSYS-01", "A11YD-01"]
- inputs_required: ["IXD-01", "DES-05", "DSYS-01", "A11YD-01", "A11YD-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the system-wide motion contract: when motion is used, what it communicates, and hard constraints to keep motion accessible, performant, and consistent. This is not a style guide for “cool animations”—it is a behavior rulebook.

Inputs Required

- IXD-01: {{xref:IXD-01}}
- DES-05: {{xref:DES-05}} | OPTIONAL
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- A11YD-05: {{xref:A11YD-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Motion principles (3–8)
- Motion use cases (minimum: navigation transitions, state changes, feedback/affordance)
- Motion constraints:
    - reduced motion policy (must)
    - duration bounds (min/max)
    - easing rules (allowed set)
    - performance constraints (avoid layout thrash, GPU-friendly guidance)
- Motion semantics mapping (what motion means)
- Prohibited motion patterns (at least 5)
- Fallback behavior when motion is disabled

Optional Fields

- Platform-specific constraints (web vs mobile) | OPTIONAL
- Component-specific motion rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Motion must always communicate one of: **relationship**, **change**, **feedback**, **focus**, **state**.
- Motion must never be the only indicator of state; pair with non-motion cues.
- Reduced-motion users must get functional equivalence (no missing affordances).
- Any motion that could induce vestibular discomfort must be disabled in reduced motion mode.
- If a rule conflicts with accessibility requirements, accessibility wins.

Output Format

1) Motion Principles (required)

- {{motion.principles[0]}}
- {{motion.principles[1]}}
- {{motion.principles[2]}}

2) Motion Semantics (required)

semantic	meaning	examples	not_used_for
relationship	{{semantics.relationship.meaning}}	{{semantics.relationship.examples}}	{{semantics.relationship.not_for}}
state_change	{{semantics.state_change.meaning}}	{{semantics.state_change.examples}}	{{semantics.state_change.not_for}}
feedback	{{semantics.feedback.meaning}}	{{semantics.feedback.examples}}	{{semantics.feedback.not_for}}
focus	{{semantics.focus.meaning}}	{{semantics.focus.examples}}	{{semantics.focus.not_for}}

3) Global Constraints (required)

- **Reduced motion policy:** {{constraints.reduced_motion.policy}}
- **Motion OFF fallback:** {{constraints.reduced_motion.fallback_behavior}}
- **Duration bounds:** min {{constraints.duration.min_ms}} ms, max {{constraints.duration.max_ms}} ms
- **Allowed easing set:** {{constraints.easing.allowed}}
- **Disallowed easing:** {{constraints.easing.disallowed}} | OPTIONAL
- **Performance rules:**
    - {{constraints.performance[0]}}
    - {{constraints.performance[1]}} | OPTIONAL

4) Motion Use Cases (required)

use_case	when_used	intent	constraints	fallback_when_reduced_motion
navigation_transition	{{use_cases.nav.when}}	{{use_cases.nav.intent}}	{{use_cases.nav.constraints}}	{{use_cases.nav.fallback}}
state_change	{{use_cases.state.when}}	{{use_cases.state.intent}}	{{use_cases.state.constraints}}	{{use_cases.state.fallback}}
feedback_affordance	{{use_cases.feedback.when}}	{{use_cases.feedback.intent}}	{{use_cases.feedback.constraints}}	{{use_cases.feedback.fallback}}

5) Prohibited Motion Patterns (required)

- {{prohibited[0]}}
- {{prohibited[1]}}
- {{prohibited[2]}}
- {{prohibited[3]}}
- {{prohibited[4]}}

6) Component/Pattern Overrides (optional)

pattern_or_component_id	override_rule	rationale
{{overrides[0].target}}	{{overrides[0].rule}}	{{overrides[0].rationale}}

Cross-References

- Upstream: {{xref:IXD-01}}, {{xref:DES-05}} | OPTIONAL
- Downstream: {{xref:IXD-03}}, {{xref:IXD-04}}, {{xref:DSYS-02}} | OPTIONAL, {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Principles + reduced motion + duration bounds + prohibited list.
- intermediate: Required. Add semantics table and use-case mapping.
- advanced: Required. Add performance constraints and overrides tied to patterns/components.

Unknown Handling

- UNKNOWN_ALLOWED: `platform_specific_constraints`, `component_overrides`, `notes`, `disallowed_easing`
- If reduced_motion policy or fallback is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IXD

- Pass conditions:
    - required_fields_present == true
    - principles_count >= 3
    - reduced_motion_defined == true
    - duration_bounds_defined == true
    - prohibited_patterns_count >= 5
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

IXD-03

IXD-03 — Transition Map (screen transitions + durations)

Header Block

- template_id: IXD-03
- title: Transition Map (screen transitions + durations)
- type: interaction_design_motion
- template_version: 1.0.0
- output_path: 10_app/design/IXD-03_Transition_Map.md
- compliance_gate_id: TMP-05.PRIMARY.IXD
- upstream_dependencies: ["DES-02", "IXD-02", "IAN-01", "IAN-02"]
- inputs_required: ["DES-02", "IXD-02", "IAN-01", "IAN-02", "DSYS-01", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical transition behavior between screens and navigation contexts. This ensures consistent mental model, supports accessibility (reduced motion), and avoids ad-hoc transition choices in implementation.

Inputs Required

- DES-02: {{xref:DES-02}}
- IAN-01: {{xref:IAN-01}} | OPTIONAL
- IAN-02: {{xref:IAN-02}} | OPTIONAL

- IXD-02: {{xref:IXD-02}}
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Navigation contexts covered (stack/tab/drawer/modal/web route changes)
- Transition types list (push/pop, modal open/close, tab switch, deep link jump, replace)
- Default transition per context
- Per-transition mapping:
    - from_screen_id
    - to_screen_id
    - transition_type
    - intent (why this transition)
    - duration_ms (must be within IXD-02 bounds)
    - reduced_motion behavior
    - interactive gesture (if any) + fallback
- Exceptions list (where defaults do not apply) with rationale

Optional Fields

- Platform-specific variants | OPTIONAL
- Notes | OPTIONAL

Rules

- Transition durations must comply with IXD-02 min/max.
- Reduced motion must have an equivalent (instant or minimal) transition.
- “Replace” navigation must not look like “back” navigation (avoid confusion).
- Modal transitions must respect focus rules and dismissal patterns (IXD-01).
- If DES-02 route is unknown, still map via screen_id; route mapping can be filled later.

Output Format

1) Navigation Context Defaults (required)

context	default_transition_type	duration_ms	reduced_motion_behavior	notes
stack_push	{{defaults.stack_push.type}}	{{defaults.stack_push.duration}}	{{defaults.stack_push.reduced_motion}}	{{defaults.stack_push.notes}}
stack_pop	{{defaults.stack_pop.type}}	{{defaults.stack_pop.duration}}	{{defaults.stack_pop.reduced_motion}}	{{defaults.stack_pop.notes}}
modal_open	{{defaults.modal_open.type}}	{{defaults.modal_open.duration}}	{{defaults.modal_open.reduced_motion}}	{{defaults.modal_open.notes}}
modal_close	{{defaults.modal_close.type}}	{{defaults.modal_close.duration}}	{{defaults.modal_close.reduced_motion}}	{{defaults.modal_close.notes}}
tab_switch	{{defaults.tab_switch.type}}	{{defaults.tab_switch.duration}}	{{defaults.tab_switch.reduced_motion}}	{{defaults.tab_switch.notes}}
deep_link	{{defaults.deep_link.type}}	{{defaults.deep_link.duration}}	{{defaults.deep_link.reduced_motion}}	{{defaults.deep_link.notes}}

2) Transition Map (required)

from_screen_id	to_screen_id	context	transition_type	intent	duration_ms	reduced_motion_behavior	gesture	gesture_fallback	notes

{{map[0].from}}	{{map[0].to}}	{{map[0].context}}	{{map[0].type}}	{{map[0].intent}}	{{map[0].duration}}	{{map[0].reduced_motion}}	{{map[0].gesture}}	{{map[0].fallback}}	{{map[0].notes}}

3) Exceptions (required if any)

exception_id	scope (context/route/screen_pair)	description	rationale	replacement_rule
ex_01	{{exceptions[0].scope}}	{{exceptions[0].description}}	{{exceptions[0].rationale}}	{{exceptions[0].replacement}}

4) Compliance Checks (required)

- All durations within bounds: {{checks.durations_within_bounds}}
- Reduced motion defined for all transitions: {{checks.reduced_motion_complete}}
- Gesture fallbacks present where gestures exist: {{checks.gesture_fallbacks_complete}}

Cross-References

- Upstream: {{xref:DES-02}}, {{xref:IXD-02}}, {{xref:IXD-01}} | OPTIONAL
- Downstream: {{xref:MOB-01}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:ROUTE-*}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define defaults per context and basic map rows.
- intermediate: Required. Add intent, reduced motion behavior, and exceptions.
- advanced: Required. Add gesture interaction and enforce compliance checks.

Unknown Handling

- UNKNOWN_ALLOWED: `platform_variants`, `notes`, `exceptions` (if none, explicitly state none)
- If any transition lacks reduced_motion_behavior → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IXD
- Pass conditions:
    - required_fields_present == true
    - defaults_present_for_all_contexts == true
    - transition_map_present == true
    - durations_within_bounds == true
    - reduced_motion_complete == true
    - gesture_fallbacks_complete == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

IXD-04

IXD-04 — Micro-interactions Spec (hover/press/drag/feedback)

Header Block

- template_id: IXD-04
- title: Micro-interactions Spec (hover/press/drag/feedback)
- type: interaction_design_motion
- template_version: 1.0.0
- output_path: 10_app/design/IXD-04_Micro_Interactions_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.IXD
- upstream_dependencies: ["IXD-01", "IXD-02", "DES-05", "DES-06", "A11YD-01"]
- inputs_required: ["IXD-01", "IXD-02", "DES-05", "DES-06", "DSYS-02", "A11YD-01", "A11YD-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the small, repeatable interaction feedback behaviors (micro-interactions) that communicate system state and affordances: hover, press, focus, drag, loading spinners, success confirmations, and subtle error feedback. These must be accessible, consistent, and implementable across platforms.

Inputs Required

- IXD-01: {{xref:IXD-01}} | OPTIONAL
- IXD-02: {{xref:IXD-02}} | OPTIONAL

- DES-05: {{xref:DES-05}} | OPTIONAL
- DES-06: {{xref:DES-06}} | OPTIONAL
- DSYS-02: {{xref:DSYS-02}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- A11YD-02: {{xref:A11YD-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Interaction types covered (minimum: hover, press/tap, focus, drag, disabled, loading, success, error)
- For each interaction:
    - interaction_id
    - target element types (button, list item, input, card, icon, etc.)
    - trigger (what user does / what system state)
    - visual feedback description (non-visual wording, no colors)
    - motion behavior (if any) + duration bounds compliance
    - haptic/audio (if mobile) + default off policy
    - accessibility behavior (focus, SR announcement if applicable)
    - do/don’t rules
    - reduced motion behavior
- Consistency rules (priority order if multiple micro-interactions apply)

Optional Fields

- Component-specific overrides | OPTIONAL
- Notes | OPTIONAL

Rules

- Micro-interactions must never be the only signal; pair with non-motion cues.
- Hover interactions must have keyboard/focus equivalents.
- Drag interactions must have a non-drag alternative (buttons, menus, etc.).
- Loading must have a maximum “no feedback” time; if longer, show progress state.
- Do not encode meaning only in color; describe feedback in state terms.

Output Format

1) Micro-interactions Catalog (canonical)

interaction_id	element_types	trigger	feedback	motion	duration_ms	reduced_motion	a11y_behavior	mobile_haptics_audio	do_dont	notes
mi_hover	{{micro[0].elements}}	{{micro[0].trigger}}	{{micro[0].feedback}}	{{micro[0].motion}}	{{micro[0].duration}}	{{micro[0].reduced_motion}}	{{micro[0].a11y}}	{{micro[0].mobile}}	{{micro[0].do_dont}}	{{micro[0].notes}}
mi_press	{{micro[1].elements}}	{{micro[1].trigger}}	{{micro[1].feedback}}	{{micro[1].motion}}	{{micro[1].duration}}	{{micro[1].reduced_motion}}	{{micro[1].a11y}}	{{micro[1].mobile}}	{{micro[1].do_dont}}	{{micro[1].notes}}

2) Detailed Rules (required)

Hover / Focus

- Hover feedback: {{rules.hover.feedback}}

- Focus feedback: {{rules.focus.feedback}}
- Keyboard equivalence: {{rules.hover.keyboard_equivalence}}

Press / Tap

- Press down: {{rules.press.down}}
- Release: {{rules.press.release}}
- Disabled press: {{rules.press.disabled}}

Drag / Reorder / Swipe

- Drag affordance: {{rules.drag.affordance}}
- Drag start threshold: {{rules.drag.threshold}} | OPTIONAL
- Drop confirmation: {{rules.drag.confirmation}}
- Non-drag fallback: {{rules.drag.fallback}}

Loading / Success / Error micro-feedback

- Loading feedback: {{rules.loading.feedback}}
- Max no-feedback time: {{rules.loading.max_no_feedback_ms}}
- Success feedback: {{rules.success.feedback}}
- Error feedback: {{rules.error.feedback}}

3) Consistency / Precedence (required)

- If disabled, disabled state overrides all: {{precedence.disabled_overrides_all}}
- If loading, loading overrides hover/press: {{precedence.loading_overrides}}
- If error, error feedback overrides success: {{precedence.error_overrides_success}}

4) Overrides (optional)

target (component_id/pattern_id)	override	rationale
{{overrides[0].target}}	{{overrides[0].override}}	{{overrides[0].rationale}}

Cross-References

- Upstream: {{xref:IXD-01}}, {{xref:IXD-02}}, {{xref:DES-05}} | OPTIONAL, {{xref:DES-06}} | OPTIONAL
- Downstream: {{xref:DSYS-02}} | OPTIONAL, {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Cover hover/press/focus/loading/error/success with clear feedback rules.
- intermediate: Required. Add drag fallbacks and precedence rules.
- advanced: Required. Add mobile haptics/audio defaults and overrides.

Unknown Handling

- UNKNOWN_ALLOWED: `mobile_haptics_audio`, `overrides`, `notes`, `drag_start_threshold`
- If max_no_feedback_ms is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IXD

- Pass conditions:
    - required_fields_present == true
    - interaction_types_minimum_covered == true
    - hover_has_focus_equivalent == true
    - drag_has_fallback == true
    - precedence_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

IXD-05

IXD-05 — Accessibility-Safe Motion Rules (reduce motion compliance)

Header Block

- template_id: IXD-05
- title: Accessibility-Safe Motion Rules (reduce motion compliance)
- type: interaction_design_motion
- template_version: 1.0.0
- output_path: 10_app/design/IXD-05_Reduce_Motion_Compliance.md
- compliance_gate_id: TMP-05.PRIMARY.IXD
- upstream_dependencies: ["IXD-02", "A11YD-01", "A11YD-05"]
- inputs_required: ["IXD-02", "IXD-03", "IXD-04", "A11YD-01", "A11YD-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the enforceable ruleset for “reduce motion” support so motion never blocks understanding or interaction. This document turns accessibility expectations into concrete constraints that can be implemented and tested.

Inputs Required

- IXD-02: {{xref:IXD-02}}
- IXD-03: {{xref:IXD-03}} | OPTIONAL
- IXD-04: {{xref:IXD-04}} | OPTIONAL

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- A11YD-05: {{xref:A11YD-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Reduce-motion policy statement (system-wide)
- Detection sources (OS setting, app toggle) and precedence
- Rules for disabling/replacing motion types:
    - parallax
    - large-scale movement
    - continuous looping animation
    - auto-play transitions
    - motion used as feedback (must provide alternate)
- Allowed motion under reduce-motion (minimal fades, instant swaps, etc.)
- Test checklist (how to validate reduce-motion)
- Exceptions policy (rare, must be justified)

Optional Fields

- Per-platform implementation notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Reduce-motion mode must preserve:
    - navigation clarity
    - state change visibility
    - feedback visibility
- Any disallowed animation must have an explicit replacement behavior.
- If app has its own toggle, it must not override OS reduce-motion by default.
- Exceptions must be documented and mapped to approver (STK).

Output Format

1) Policy (required)

- **Policy:** {{reduce_motion.policy}}
- **Applies to:** {{reduce_motion.applies_to}} (all users / only when enabled)
- **Precedence:** {{reduce_motion.precedence}} (OS setting > app toggle, etc.)

2) Detection (required)

source	how_detected	precedence	notes
os_setting	{{detect.os.how}}	1	{{detect.os.notes}}
app_toggle	{{detect.app.how}}	2	{{detect.app.notes}}

3) Disallowed Motion Types (required)

motion_type	disallowed_behavior	replacement_behavior	rationale
parallax	{{disallowed.parallax.behavior}}	{{disallowed.parallax.replacement}}	{{disallowed.parallax.rationale}}

large_scale_movement	{{disallowed.large.behavior}}	{{disallowed.large.replacement}}	{{disallowed.large.rationale}}
looping_animation	{{disallowed.looping.behavior}}	{{disallowed.looping.replacement}}	{{disallowed.looping.rationale}}
auto_play_transition	{{disallowed.auto.behavior}}	{{disallowed.auto.replacement}}	{{disallowed.auto.rationale}}
motion_feedback	{{disallowed.feedback.behavior}}	{{disallowed.feedback.replacement}}	{{disallowed.feedback.rationale}}

4) Allowed Motion (required)

- Allowed minimal transitions: {{allowed.minimal_transitions}}
- Duration cap in reduce-motion: {{allowed.duration_cap_ms}}
- Allowed easing set: {{allowed.easing}}

5) Exceptions Policy (required)

- When exceptions are allowed: {{exceptions.when_allowed}}
- Required justification fields: {{exceptions.justification_fields}}
- Required approver: {{exceptions.approver_stakeholder_id}} | OPTIONAL
- Logging requirement: {{exceptions.logging}} | OPTIONAL

6) Test Checklist (required)

- {{tests[0]}}
- {{tests[1]}}
- {{tests[2]}}
- {{tests[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:IXD-02}}, {{xref:A11YD-01}} | OPTIONAL
- Downstream: {{xref:QA-02}} | OPTIONAL, {{xref:RJT-*}} | OPTIONAL, {{xref:DSYS-01}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Policy + disallowed list + replacements + test checklist.
- intermediate: Required. Add detection precedence and allowed motion constraints.
- advanced: Required. Add exceptions governance and logging requirements.

Unknown Handling

- UNKNOWN_ALLOWED: `per_platform_notes`, `exceptions.approver`, `exceptions.logging`, `notes`
- If any disallowed motion type lacks replacement_behavior → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IXD
- Pass conditions:
    - required_fields_present == true
    - policy_defined == true
    - detection_defined == true
    - all_disallowed_have_replacements == true
    - tests_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

Content Design & UX Writing (CDX)

Content Design & UX Writing (CDX)

- CDX-01 Content Style Guide (tone, voice, terminology)
- CDX-02 UI Copy Inventory (labels, buttons, helper text)
- CDX-03 Empty States & Onboarding Copy
- CDX-04 Error/Warning/Success Message Catalog
- CDX-05 Notification Copy Templates (push/email/in-app)

CDX-01

CDX-01 — Content Style Guide (tone, voice, terminology)

Header Block

- template_id: CDX-01
- title: Content Style Guide (tone, voice, terminology)
- type: content_design_ux_writing
- template_version: 1.0.0
- output_path: 10_app/content/CDX-01_Content_Style_Guide.md
- compliance_gate_id: TMP-05.PRIMARY.CONTENT
- upstream_dependencies: ["DMG-01", "PRD-03", "PRD-01"]
- inputs_required: ["DMG-01", "PRD-01", "PRD-03", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical content rules the product uses everywhere (UI copy, onboarding, notifications, errors): voice, tone, terminology, readability, and accessibility-safe writing. This prevents copy drift and makes content production deterministic.

Inputs Required

- DMG-01: {{xref:DMG-01}} | OPTIONAL
- PRD-01: {{xref:PRD-01}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Brand voice definition (3–8 traits)
- Tone rules by context (success, neutral, error, warning, onboarding, empty state)
- Terminology rules:
    - canonical terms (from DMG-01)
    - forbidden / deprecated terms
    - naming rules for roles, features, plans (if applicable)
- Readability rules (grade level or simplicity rules)
- Inclusive language rules
- Localization readiness rules (writing to translate well)
- Accessibility writing rules (screen reader-friendly text patterns)
- Examples: “do/don’t” pairs (minimum 10)

Optional Fields

- Legal/compliance wording constraints | OPTIONAL
- Brand punctuation/casing preferences | OPTIONAL
- Notes | OPTIONAL

Rules

- Canonical terms must match DMG-01; if conflict, update glossary or log decision.
- Error copy must be actionable and non-blaming.
- Avoid idioms and culture-specific slang if localization is expected.
- Do not encode meaning in emoji alone; if emojis are used, they are decorative and optional.
- Messages should be scannable: front-load key info; keep sentences short.

Output Format

1) Voice (required)

- **Voice traits:** {{voice.traits}}
- **What we sound like:** {{voice.description}}
- **What we never sound like:** {{voice.never}} | OPTIONAL

2) Tone by Context (required)

context	tone	goal	example
success	{{tone.success.tone}}	{{tone.success.goal}}	{{tone.success.example}}
neutral	{{tone.neutral.tone}}	{{tone.neutral.goal}}	{{tone.neutral.example}}
warning	{{tone.warning.tone}}	{{tone.warning.goal}}	{{tone.warning.example}}
error	{{tone.error.tone}}	{{tone.error.goal}}	{{tone.error.example}}
onboarding	{{tone.onboarding.tone}}	{{tone.onboarding.goal}}	{{tone.onboarding.example}}
empty_state	{{tone.empty.tone}}	{{tone.empty.goal}}	{{tone.empty.example}}

3) Terminology Rules (required)

- **Canonical terms source:** {{xref:DMG-01}} | OPTIONAL

- **Role naming rules:** {{terminology.roles}}
- **Feature naming rules:** {{terminology.features}} | OPTIONAL
- **Plan/tier naming rules:** {{terminology.plans}} | OPTIONAL
- **Forbidden/deprecated terms:** {{terminology.forbidden}} | OPTIONAL

4) Readability & Structure (required)

- Sentence length guidance: {{readability.sentence_length}}
- Reading level target: {{readability.level}}
- Scannability rules (bullets, headings): {{readability.scannability}}

5) Inclusive Language (required)

- {{inclusive[0]}}
- {{inclusive[1]}} | OPTIONAL

6) Localization Readiness (required)

- Avoid: idioms, jokes, region-specific formats unless localized: {{l10n.avoid}}
- Date/time/number placeholders rules: {{l10n.placeholders}}
- Gendered language policy: {{l10n.gender}} | OPTIONAL

7) Accessibility Writing (required)

- Screen reader clarity rules: {{a11y.text_rules}}
- Link text rules (no “click here”): {{a11y.links}}
- Error message structure (what happened / what to do): {{a11y.error_structure}}

8) Do/Don’t Examples (required, min 10)

example_id	do	dont	why
ex_01	{{examples[0].do}}	{{examples[0].dont}}	{{examples[0].why}}
ex_02	{{examples[1].do}}	{{examples[1].dont}}	{{examples[1].why}}

9) Legal/Compliance Constraints (optional)

- {{legal[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-01}} | OPTIONAL, {{xref:PRD-01}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL
- Downstream: {{xref:CDX-02}}, {{xref:CDX-03}}, {{xref:CDX-04}}, {{xref:CDX-05}}
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Voice + tone table + 10 do/don't examples.
- intermediate: Required. Add terminology and localization rules.
- advanced: Required. Add accessibility writing patterns and legal constraints if needed.

Unknown Handling

- UNKNOWN_ALLOWED: `legal_constraints`, `punctuation_preferences`, `notes`, `forbidden_terms`
- If tone rules are UNKNOWN for error or warning contexts → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CONTENT
- Pass conditions:
    - required_fields_present == true
    - voice_traits_present == true
    - tone_by_context_present == true
    - do_dont_examples_count >= 10
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CDX-02

CDX-02 — UI Copy Inventory (labels, buttons, helper text)

Header Block

- template_id: CDX-02
- title: UI Copy Inventory (labels, buttons, helper text)
- type: content_design_ux_writing
- template_version: 1.0.0
- output_path: 10_app/content/CDX-02_UI_Copy_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.CONTENT
- upstream_dependencies: ["DES-02", "DES-03", "CDX-01"]
- inputs_required: ["DES-02", "DES-03", "CDX-01", "DMG-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical inventory of UI copy strings used across screens: titles, labels, buttons, helper text, tooltips, and confirmations. This enables consistency, localization, and implementation without copy drift.

Inputs Required

- DES-02: {{xref:DES-02}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL
- CDX-01: {{xref:CDX-01}}

- DMG-01: {{xref:DMG-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Copy entries list (minimum 50 for non-trivial products; justify if smaller)
- For each entry:
    - copy_id
    - screen_id (or global)
    - component_id (optional)
    - location (header, field label, button, tooltip, helper text)
    - string_key (implementation key)
    - default_text
    - intent (what it communicates)
    - tone context (from CDX-01)
    - placeholders (if any)
    - accessibility note (if SR/aria label differs)
    - max length guidance (if constrained)
    - localization note (if special)
- Canonical naming rules for string_key

Optional Fields

- Variants by platform (web/mobile) | OPTIONAL

- Notes | OPTIONAL

Rules

- Must comply with CDX-01 voice/tone/terminology.
- Placeholders must be explicit and consistent (e.g., {name}, {count}).
- Avoid embedding PII in logs; copy entries should not require showing sensitive values unless allowed.
- If the visible label differs from the aria label, both must be specified.

Output Format

1) String Key Rules (required)

- Key format: {{keys.format}} (e.g., screen.<screen_id>.<location>.<name>)
- Allowed characters: {{keys.allowed_chars}}
- Versioning/change rules: {{keys.change_rules}} | OPTIONAL

2) UI Copy Inventory (canonical)

copy_id	string_key	screen_id	component_id	location	default_text	intent	tone_context	placeholders	a11y_alt_text	max_length	l10n_notes	notes
c_001	{{copy[0].key}}	{{copy[0].screen_id}}	{{copy[0].component_id}}	{{copy[0].location}}	{{copy[0].text}}	{{copy[0].intent}}	{{copy[0].tone}}	{{copy[0].placeholders}}	{{copy[0].a11y_alt}}	{{copy[0].max_len}}	{{copy[0].l10n}}	{{copy[0].notes}}

c_002	{{copy[1].key}}	{{copy[1].screen_id}}	{{copy[1].component_id}}	{{copy[1].location}}	{{copy[1].text}}	{{copy[1].intent}}	{{copy[1].tone}}	{{copy[1].placeholders}}	{{copy[1].a11y_alt}}	{{copy[1].max_len}}	{{copy[1].l10n}}	{{copy[1].notes}}

3) Coverage Checks (required)

- Screens covered: {{coverage.screens}} | OPTIONAL
- Global copy present (app name, generic CTAs): {{coverage.global}}
- A11y alt text present where needed: {{coverage.a11y_alt}}

Cross-References

- Upstream: {{xref:CDX-01}}, {{xref:DES-02}} | OPTIONAL, {{xref:DES-03}} | OPTIONAL
- Downstream: {{xref:CDX-03}}, {{xref:CDX-04}}, {{xref:CDX-05}} | OPTIONAL, {{xref:FE-*}} | *OPTIONAL, {{xref:L10N-}}* | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Inventory with keys + default text + screen mapping.
- intermediate: Required. Add placeholders, max length guidance, and intent.
- advanced: Required. Add a11y alt labels and localization notes.

Unknown Handling

- UNKNOWN_ALLOWED: `component_id`, `variants_by_platform`, `max_length`, `l10n_notes`, `notes`, `a11y_alt_text`

- If string_key rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CONTENT
- Pass conditions:
    - required_fields_present == true
    - copy_entries_count >= 50 (or justified)
    - string_keys_unique == true
    - cdX01_compliance == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CDX-03

CDX-03 — Empty States & Onboarding Copy

Header Block

- template_id: CDX-03
- title: Empty States & Onboarding Copy
- type: content_design_ux_writing
- template_version: 1.0.0
- output_path: 10_app/content/CDX-03_EmptyStates_Onboarding_Copy.md
- compliance_gate_id: TMP-05.PRIMARY.CONTENT
- upstream_dependencies: ["DES-05", "DES-03", "CDX-01", "CDX-02"]
- inputs_required: ["DES-05", "DES-03", "CDX-01", "CDX-02", "URD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical copy for empty states and onboarding moments so the product consistently guides users when there's nothing to show yet, or when users are learning the system. This includes messaging, CTAs, and intent—mapped to screens and states.

Inputs Required

- DES-05: {{xref:DES-05}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL
- CDX-01: {{xref:CDX-01}}

- CDX-02: {{xref:CDX-02}} | OPTIONAL
- URD-03: {{xref:URD-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Empty state entries (minimum 10)
- Onboarding entries (minimum 5)
- For each entry:
    - copy_id
    - screen_id
    - state (empty/onboarding/first_run/no_results/no_access)
    - title text
    - body text
    - CTA text (if any)
    - CTA action intent (what it does)
    - eligibility/access condition (if relevant)
    - linked feature_id(s) (optional)
    - tone context (from CDX-01)
    - accessibility notes (SR-friendly wording)
    - localization notes (if any)

Optional Fields

- Visual companion guidance (what an illustration should convey) | OPTIONAL
- Progressive onboarding sequence (steps) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must comply with CDX-01 voice/tone rules.
- Empty states must be actionable where possible (suggest next step).
- Never blame the user; keep language constructive.
- If state is “no access,” language must align to entitlements and avoid exposing restricted details.
- Keep onboarding steps short; one concept per step.

Output Format

1) Empty States Catalog (required)

copy_id	screen_id	state_type	title	body	cta_text	cta_intent	access_condition	feature_ids	tone	a11y_notes	l10n_notes	notes
es_001	{{empty[0].screen_id}}	{{empty[0].state_type}}	{{empty[0].title}}	{{empty[0].body}}	{{empty[0].cta}}	{{empty[0].cta_intent}}	{{empty[0].access}}	{{empty[0].feature_ids}}	{{empty[0].tone}}	{{empty[0].a11y}}	{{empty[0].l10n}}	{{empty[0].notes}}
es_	{{empty[1].sc	{{empty[1].st	{{empty[	{{empty[1	{{empty[1].	{{empty[1].ct	{{empty[1]	{{empty[1].fe	{{empty[1	{{empty[1	{{empty[1	{{empty[1

002	reen_id}}	ate_type}}	1].title}}	].body}}	cta}}	a_intent}}	.access}}	ature_ids}}	].tone}}	].a11y}}	].l10n}}	].notes}}

2) Onboarding Copy Catalog (required)

copy_id	screen_id	onboarding_step	title	body	cta_text	cta_intent	feature_ids	tone	a11y_notes	l10n_notes	notes
ob_001	{{onboarding[0].screen_id}}	{{onboarding[0].step}}	{{onboarding[0].title}}	{{onboarding[0].body}}	{{onboarding[0].cta}}	{{onboarding[0].cta_intent}}	{{onboarding[0].feature_ids}}	{{onboarding[0].tone}}	{{onboarding[0].a11y}}	{{onboarding[0].l10n}}	{{onboarding[0].notes}}

3) Coverage Checks (required)

- Empty states cover all screens with empty state in DES-05: {{coverage.empty_states_complete}}
- Onboarding exists for primary first-run flows: {{coverage.onboarding_complete}}

4) Visual Companion Guidance (optional)

copy_id	concept_to_convey	do_not_convey
{{visuals[0].copy_id}}	{{visuals[0].concept}}	{{visuals[0].avoid}}

Cross-References

- Upstream: {{xref:CDX-01}}, {{xref:CDX-02}} | OPTIONAL, {{xref:DES-05}} | OPTIONAL, {{xref:URD-03}} | OPTIONAL
- Downstream: {{xref:VAP-01}} | OPTIONAL, {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL

- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 10 empty states + 5 onboarding entries with titles/bodies/CTAs.
- intermediate: Required. Add access conditions and CTA intent mapping.
- advanced: Required. Add a11y/l10n notes and visual companion guidance.

Unknown Handling

- UNKNOWN_ALLOWED: `feature_ids`, `visual_companion_guidance`, `progressive_sequence`, `notes`, `l10n_notes`
- If an empty state has a CTA, cta_intent cannot be UNKNOWN.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CONTENT
- Pass conditions:
  - required_fields_present == true
  - empty_entries_count >= 10
  - onboarding_entries_count >= 5
  - coverage_checks_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

CDX-04

CDX-04 — Error/Warning/Success Message Catalog

Header Block

- template_id: CDX-04
- title: Error/Warning/Success Message Catalog
- type: content_design_ux_writing
- template_version: 1.0.0
- output_path: 10_app/content/CDX-04_Message_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.CONTENT
- upstream_dependencies: ["DES-07", "DES-06", "ARC-06", "API-03", "CDX-01"]
- inputs_required: ["DES-07", "DES-06", "CDX-01", "ARC-06", "API-03", "A11YD-05", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical catalog of user-facing messages for success, warning, and error states. This ensures consistent wording, supports reason-code mapping, and prevents sensitive detail leakage.

Inputs Required

- DES-07: {{xref:DES-07}} | OPTIONAL
- DES-06: {{xref:DES-06}} | OPTIONAL
- CDX-01: {{xref:CDX-01}}

- ARC-06: {{xref:ARC-06}} | OPTIONAL
- API-03: {{xref:API-03}} | OPTIONAL
- A11YD-05: {{xref:A11YD-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Message entries (minimum 30 for non-trivial products; justify if smaller)
- For each message:
    - msg_id
    - category (success/warning/error/info)
    - trigger condition (what causes it)
    - surface (inline/toast/banner/modal)
    - title (optional)
    - body text (required)
    - user action guidance (what user can do next)
    - severity (P0/P1/P2)
    - reason_code mapping (optional: rc_*)
    - retry_allowed (true/false)
    - accessibility notes (announcement behavior)
    - localization notes
    - redaction rule (what must not be shown)

Optional Fields

- Variants by platform | OPTIONAL
- Debug correlation ID policy (display or not) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must comply with CDX-01 voice/tone rules.
- Error messages must not leak internal system details; use redaction rules.
- If mapped to reason codes (ARC-06), the mapping must be explicit.
- Every P0 error must include user action guidance.
- If retry_allowed is true, the message must align to DES-07 retry rules.

Output Format

1) Message Catalog (canonical)

msg_id	category	surface	trigger	title	body	user_guidance	severity	reason_code	retry_allowed	a11y_notes	l10n_notes	redaction_rule	notes
msg_001	{{msgs[0].category}}	{{msgs[0].surface}}	{{msgs[0].trigger}}	{{msgs[0].title}}	{{msgs[0].body}}	{{msgs[0].guidance}}	{{msgs[0].severity}}	{{msgs[0].reason_code}}	{{msgs[0].retry_allowed}}	{{msgs[0].a11y}}	{{msgs[0].l10n}}	{{msgs[0].redaction}}	{{msgs[0].notes}}
msg_	{{msgs[1].cate	{{msgs[1].sur	{{msgs[1].trig	{{msgs[1].t	{{msgs[1].b	{{msgs[1].guid	{{msgs[1].sev	{{msgs[1].reason_code}}	{{msgs[1].retry_allowed}}	{{msgs[1].a	{{msgs[1].l	{{msgs[1].reda	{{msgs[1].n

0	gory}	face	ger}	itle}	ody	ance	erity}	11y	10n	ction}	otes
0	}	}}	}	}	}}	}}	}	}}	}}	}	}}
2											

2) Coverage Checks (required)

- P0 errors have guidance: {{coverage.p0_errors_have_guidance}}
- Retry-allowed messages align with DES-07: {{coverage.retry_alignment}}
- Reason-code mappings (if used) are valid: {{coverage.reason_codes_valid}}

3) Redaction Rules (required)

- Never show: {{redaction.never_show}}
- Allowed to show: {{redaction.allowed}} | OPTIONAL
- Correlation ID policy: {{redaction.correlation_id_policy}} | OPTIONAL

Cross-References

- Upstream: {{xref:CDX-01}}, {{xref:DES-07}} | OPTIONAL, {{xref:ARC-06}} | OPTIONAL, {{xref:API-03}} | OPTIONAL
- Downstream: {{xref:DES-05}} | OPTIONAL, {{xref:FE-07}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 30 messages with body + surface + guidance for P0.
- intermediate: Required. Add retry flags and redaction rules.

- advanced: Required. Add reason-code mapping and a11y announcement notes.

Unknown Handling

- UNKNOWN_ALLOWED: `title`, `reason_code`, `variants_by_platform`, `correlation_id_policy`, `notes`, `l10n_notes`
- If severity == P0 and user_guidance is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CONTENT
- Pass conditions:
    - required_fields_present == true
    - messages_count >= 30 (or justified)
    - p0_errors_have_guidance == true
    - redaction_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

CDX-05

CDX-05 — Notification Copy Templates (push/email/in-app)

Header Block

- template_id: CDX-05
- title: Notification Copy Templates (push/email/in-app)
- type: content_design_ux_writing
- template_version: 1.0.0
- output_path: 10_app/content/CDX-05_Notification_Copy_Templates.md
- compliance_gate_id: TMP-05.PRIMARY.CONTENT
- upstream_dependencies: ["CDX-01", "CDX-02", "CDX-04", "DMG-04"]
- inputs_required: ["CDX-01", "CDX-02", "CDX-04", "DMG-04", "MSG-03", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define canonical, reusable copy templates for notifications across channels (push, email, in-app). This ensures consistent tone, predictable personalization, safe handling of sensitive data, and deterministic mapping from events to messages.

Inputs Required

- CDX-01: {{xref:CDX-01}}
- CDX-02: {{xref:CDX-02}} | OPTIONAL
- CDX-04: {{xref:CDX-04}} | OPTIONAL

- DMG-04: {{xref:DMG-04}} | OPTIONAL
- MSG-03: {{xref:MSG-03}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Channel rules (push/email/in-app) including length constraints and required parts
- Template list (minimum 15; justify if smaller)
- For each template:
    - notif_id
    - category (transactional/engagement/security/system)
    - trigger event(s) (event_id/event_name)
    - audience/eligibility (role/tier/segment) (optional)
    - channel(s) (push/email/in-app)
    - title/subject (per channel where applicable)
    - body content (per channel)
    - CTA text + CTA intent (where applicable)
    - personalization placeholders (explicit list)
    - redaction rules (PII / sensitive constraints)
    - dedupe key / frequency cap note (copy-side expectations)
    - localization notes
    - accessibility notes (readable, SR-friendly)

Optional Fields

- Variants (A/B copy options) | OPTIONAL
- Quiet hours policy note (pointer) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must comply with CDX-01 voice/tone and terminology.
- Must comply with privacy classification (DGP-01): no high-PII values in push content unless explicitly allowed.
- Push notifications must be understandable without opening the app.
- Email templates must include a clear subject and a single primary CTA (unless system-critical).
- Any security-related notification must avoid revealing exploitable details (e.g., “wrong password” specifics).
- Placeholders must be explicit and safe (no raw PII unless allowed by DGP).

Output Format

1) Channel Constraints (required)

channel	must_include	max_length_guidance	tone_context	notes
push	{{channels.push.must_include}}	{{channels.push.max_len}}	{{channels.push.tone}}	{{channels.push.notes}}
in_app	{{channels.in_app.must_include}}	{{channels.in_app.max_len}}	{{channels.in_app.tone}}	{{channels.in_app.notes}}
email	{{channels.email.must_include}}	{{channels.email.max_len}}	{{channels.email.tone}}	{{channels.email.notes}}

2) Notification Templates (canonical)

notif_id	category	trigger_events	channels	push_title	push_body	in_app_title	in_app_body	email_subject	email_body	cta_text	cta_intent	placeholders	reduction_rules	freq_cap_dedup_note	a11y_notes	l10n_notes	notes
notif_001	{{templates[0].category}}	{{templates[0].events}}	{{templates[0].channels}}	{{templates[0].push_title}}	{{templates[0].push_body}}	{{templates[0].in_app_title}}	{{templates[0].in_app_body}}	{{templates[0].email_subject}}	{{templates[0].email_body}}	{{templates[0].cta_text}}	{{templates[0].cta_intent}}	{{templates[0].placeholders}}	{{templates[0].reduction}}	{{templates[0].freq_cap}}	{{templates[0].a11y}}	{{templates[0].l10n}}	{{templates[0].notes}}

3) Placeholder Policy (required)

- Allowed placeholders: {{placeholders.allowed}}
- Disallowed placeholders (PII/sensitive): {{placeholders.disallowed}}
- Formatting rules (dates, counts): {{placeholders.formatting}}

4) Privacy/Redaction Rules (required)

- Never include in push: {{privacy.never_in_push}}
- Allowed in email (if consented): {{privacy.allowed_in_email}} | OPTIONAL
- Security notification restrictions: {{privacy.security_notification_rules}}

5) Coverage Checks (required)

- Triggers map to known events (DMG-04): {{coverage.events_valid}}
- Each template has channel-appropriate fields: {{coverage.channel_fields_complete}}
- No forbidden placeholders used: {{coverage.no_forbidden_placeholders}}

Cross-References

- Upstream: {{xref:CDX-01}}, {{xref:DMG-04}} | OPTIONAL, {{xref:MSG-03}} | OPTIONAL, {{xref:DGP-01}} | OPTIONAL
- Downstream: {{xref:NOTIF-*}} | OPTIONAL, {{xref:INT-03}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 15 templates with push+in-app basics and clear CTAs.
- intermediate: Required. Add placeholder lists, redaction rules, and event mapping.
- advanced: Required. Add channel constraints, freqcap/dedupe notes, and localization guidance.

Unknown Handling

- UNKNOWN_ALLOWED: `variants`, `quiet_hours_policy_note`, `freqcap_dedupe_note`, `l10n_notes`, `notes`
- If a template has a CTA, cta_intent cannot be UNKNOWN.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.CONTENT

- Pass conditions:
    - required_fields_present == true
    - templates_count >= 15 (or justified)
    - events_valid == true
    - channel_fields_complete == true
    - no_forbidden_placeholders == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Design System & UI Tokens (DSYS)

Design System & UI Tokens (DSYS)

- DSYS-01 Token Spec (color, type, spacing, radius, elevation)
- DSYS-02 Component Variants Spec (props, variants, states)
- DSYS-03 Layout Grid & Spacing Rules
- DSYS-04 Iconography & Illustration Rules
- DSYS-05 Theming Rules (light/dark, brand constraints)

DSYS-01

DSYS-01 — Token Spec (color, type, spacing, radius, elevation)

Header Block

- template_id: DSYS-01
- title: Token Spec (color, type, spacing, radius, elevation)
- type: design_system_tokens
- template_version: 1.0.0
- output_path: 10_app/design_system/DSYS-01_Token_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.DSYS
- upstream_dependencies: ["CDX-01", "A11YD-04"]
- inputs_required: ["CDX-01", "A11YD-04", "RLB-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical design tokens used across UI implementation so styling is consistent, themeable, and accessible. Tokens are the source of truth for UI values (not component rules), enabling FE/MOB to implement without inventing new visual constants.

Inputs Required

- CDX-01: {{xref:CDX-01}} | OPTIONAL
- A11YD-04: {{xref:A11YD-04}} | OPTIONAL
- RLB-01: {{xref:RLB-01}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing brand palette: {{inputs.brand_palette}} | OPTIONAL

Required Fields

- Token namespaces (color/type/space/radius/elevation/border/shadow/zindex)
- Color tokens:
    - semantic roles (bg/surface/text/border/primary/success/warn/error)
    - states (hover/active/disabled/focus)
    - theme variants (light/dark if applicable)
- Typography tokens (font families, sizes, weights, line heights)
- Spacing scale tokens (consistent step scale)
- Radius tokens (corner radii scale)
- Elevation tokens (shadows/surfaces) OR depth scale definition
- Focus tokens (focus ring width/offset/semantic color)
- Token naming rules and stability rules

Optional Fields

- Motion tokens (durations/easing) | OPTIONAL (may reference IXD)
- Component alias tokens (e.g., button padding) | OPTIONAL
- Notes | OPTIONAL

Rules

- Tokens must be semantic-first (meaning-based), not “random values.”
- Token names must be stable; changes are versioned and should avoid breaking downstream.
- All text and interactive tokens must meet contrast requirements (A11YD-04).
- If multiple themes exist, every semantic token must have values in each theme.
- No “magic numbers” in implementation: all UI values should reference tokens unless explicitly exempt.

Output Format

1) Token Naming Rules (required)

- Namespace format: {{tokens.naming.namespace_format}} (e.g., color.bg.default)
- Allowed characters: {{tokens.naming.allowed_chars}}
- Aliasing rules: {{tokens.naming.aliasing_rules}}
- Deprecation rules: {{tokens.naming.deprecation_rules}} | OPTIONAL

2) Color Tokens (required)

token	semantic_role	light_value	dark_value	states	contrast_notes	usage_notes
{{colors[0].token}}	{{colors[0].role}}	{{colors[0].light}}	{{colors[0].dark}}	{{colors[0].states}}	{{colors[0].contrast}}	{{colors[0].usage}}
{{colors[1].token}}	{{colors[1].role}}	{{colors[1].light}}	{{colors[1].dark}}	{{colors[1].states}}	{{colors[1].contrast}}	{{colors[1].usage}}

3) Typography Tokens (required)

token	font_family	size	weight	line_height	letter_spacing	usage_notes
{{type[0].token}}	{{type[0].family}}	{{type[0].size}}	{{type[0].weight}}	{{type[0].line_height}}	{{type[0].letter_spacing}}	{{type[0].usage}}

4) Spacing Scale (required)

token	value	usage_notes
{{space[0].token}}	{{space[0].value}}	{{space[0].usage}}

5) Radius Scale (required)

token	value	usage_notes
{{radius[0].token}}	{{radius[0].value}}	{{radius[0].usage}}

6) Elevation / Depth Scale (required)

token	description	value	usage_notes
{{elevation[0].token}}	{{elevation[0].desc}}	{{elevation[0].value}}	{{elevation[0].usage}}

7) Focus Tokens (required)

token	value	usage_notes
{{focus[0].token}}	{{focus[0].value}}	{{focus[0].usage}}

8) Theme Coverage Check (required)

- Has light theme: {{themes.has_light}}
- Has dark theme: {{themes.has_dark}} | OPTIONAL
- All semantic tokens covered in each theme: {{themes.coverage_complete}}
- Contrast checks complete (A11YD-04): {{themes.contrast_complete}}

Cross-References

- Upstream: {{xref:A11YD-04}} | OPTIONAL, {{xref:RLB-01}} | OPTIONAL
- Downstream: {{xref:DSYS-02}}, {{xref:FE-06}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define core semantic tokens (primary/bg/text/border/status) + spacing/radius.
- intermediate: Required. Add typography and elevation; define naming rules.
- advanced: Required. Add theme coverage + focus tokens + contrast verification notes.

Unknown Handling

- UNKNOWN_ALLOWED: `motion_tokens`, `component_alias_tokens`, `notes`, `dark_theme_values` (if only light theme)
- If contrast_notes are UNKNOWN for text tokens → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DSYS
- Pass conditions:
    - required_fields_present == true
    - naming_rules_present == true
    - color_tokens_present == true
    - typography_tokens_present == true
    - spacing_radius_elevation_present == true
    - theme_coverage_complete == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DSYS-02

DSYS-02 — Component Variants Spec (props, variants, states)

Header Block

- template_id: DSYS-02
- title: Component Variants Spec (props, variants, states)
- type: design_system_tokens
- template_version: 1.0.0
- output_path: 10_app/design_system/DSYS-02_Component_Variants_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.DSYS
- upstream_dependencies: ["DSYS-01", "DES-04", "A11YD-01"]
- inputs_required: ["DSYS-01", "DES-04", "A11YD-01", "A11YD-02", "A11YD-05", "IXD-01", "IXD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the design-system component contract: which components exist, what variants they support, what states they must handle, and what props/inputs they accept at a conceptual level. This enables FE/MOB to implement consistently without inventing component behavior.

Inputs Required

- DSYS-01: {{xref:DSYS-01}}
- DES-04: {{xref:DES-04}} | OPTIONAL
- IXD-01: {{xref:IXD-01}} | OPTIONAL

- IXD-04: {{xref:IXD-04}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- A11YD-02: {{xref:A11YD-02}} | OPTIONAL
- A11YD-05: {{xref:A11YD-05}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Component list (minimum 12 for non-trivial products)
- For each component:
    - dsys_component_id (or component_id alignment to DES-04)
    - name
    - purpose
    - supported variants (e.g., size, style, intent)
    - required states (default/hover/active/focus/disabled/loading/error)
    - props/inputs (conceptual; name + meaning)
    - output events (onClick/onChange/etc.)
    - accessibility contract (keyboard, focus, labels, SR semantics)
    - content rules (label length, wrapping, truncation policy)
    - motion hooks (if any; reference IXD)
- Variant consistency rules (what “primary/secondary/destructive” means across components)
- State precedence rules (disabled overrides hover, etc.)

Optional Fields

- Platform-specific differences (web/mobile) | OPTIONAL
- Deprecation/compat notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Each component must declare its required states; missing states are not allowed.
- “Focus” behavior must be explicit and align with A11Y focus specs.
- Variants must be semantic (intent-driven), not “random style names.”
- Any motion hooks must respect IXD reduce-motion rules.
- Props must be stable; changes require versioning notes.

Output Format

1) Variant Semantics (required)

Define shared meaning for common variant names.

variant_name	meaning	do	dont
primary	{{variant_semantics.primary.meaning}}	{{variant_semantics.primary.do}}	{{variant_semantics.primary.dont}}
secondary	{{variant_semantics.secondary.meaning}}	{{variant_semantics.secondary.do}}	{{variant_semantics.secondary.dont}}
destructive	{{variant_semantics.destructive.meaning}}	{{variant_semantics.destructive.do}}	{{variant_semantics.destructive.dont}}

2) State Precedence Rules (required)

- disabled overrides hover/active/focus: {{state_precedence.disabled}}
- loading overrides interaction (unless cancel): {{state_precedence.loading}}
- error overrides success feedback: {{state_precedence.error}} | OPTIONAL

3) Component Contract Catalog (canonical)

component_id	name	purpose	variants	required_states	props_inputs	output_events	a11y_contract	content_rules	motion_hooks	platform_notes	notes
{{components[0].id}}	{{components[0].name}}	{{components[0].purpose}}	{{components[0].variants}}	{{components[0].states}}	{{components[0].props}}	{{components[0].events}}	{{components[0].a11y}}	{{components[0].content}}	{{components[0].motion}}	{{components[0].platform}}	{{components[0].notes}}
{{components[1].id}}	{{components[1].name}}	{{components[1].purpose}}	{{components[1].variants}}	{{components[1].states}}	{{components[1].props}}	{{components[1].events}}	{{components[1].a11y}}	{{components[1].content}}	{{components[1].motion}}	{{components[1].platform}}	{{components[1].notes}}

4) Coverage Checks (required)

- Components cover DES-04 inventory: {{coverage.covers_des04}} | OPTIONAL
- Required states present for all components: {{coverage.states_complete}}
- A11y contract present for all interactive components: {{coverage.a11y_complete}}

Cross-References

- Upstream: {{xref:DSYS-01}}, {{xref:DES-04}} | OPTIONAL

- Downstream: {{xref:FE-02}} | OPTIONAL, {{xref:FE-06}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define component list + variants + required states.
- intermediate: Required. Add props/events and state precedence.
- advanced: Required. Add a11y contracts, content rules, and motion hooks.

Unknown Handling

- UNKNOWN_ALLOWED: `platform_notes`, `deprecation_notes`, `notes`, `motion_hooks` (if none)
- If any interactive component lacks a11y_contract → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DSYS
- Pass conditions:
    - required_fields_present == true
    - components_count >= 12
    - required_states_complete == true
    - a11y_contract_complete == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DSYS-03

DSYS-03 — Layout Grid & Spacing Rules

Header Block

- template_id: DSYS-03
- title: Layout Grid & Spacing Rules
- type: design_system_tokens
- template_version: 1.0.0
- output_path: 10_app/design_system/DSYS-03_Layout_Grid_Spacing_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.DSYS
- upstream_dependencies: ["DSYS-01", "RLB-01"]
- inputs_required: ["DSYS-01", "RLB-01", "RLB-02", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the system-wide layout and spacing rules so screens and components align to a consistent grid and density model across breakpoints. This prevents arbitrary spacing and ensures predictable responsive behavior.

Inputs Required

- DSYS-01: {{xref:DSYS-01}}
- RLB-01: {{xref:RLB-01}} | OPTIONAL
- RLB-02: {{xref:RLB-02}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Grid model definition (web and/or mobile)
    - columns (if applicable)
    - gutters
    - margins/containers
    - baseline spacing unit
- Spacing scale usage rules (tie to DSYS-01 spacing tokens)
- Density rules (comfortable/compact) if applicable
- Alignment rules (edges, baselines, centers) and when to use each
- Common layout patterns (list/detail, cards, forms, dashboards)
- Touch target spacing rules (must align with A11Y/touch guidance)
- Exceptions policy (when breaking grid is allowed)

Optional Fields

- Platform differences (native vs web) | OPTIONAL
- Layout anti-patterns | OPTIONAL
- Notes | OPTIONAL

Rules

- Spacing must use tokens; no ad-hoc values unless explicitly exempt.

- Any exception must be justified and scoped (screen/component).
- Touch targets must remain accessible across density modes.
- Responsive behavior must not change information hierarchy unintentionally.

Output Format

1) Grid Definition (required)

platform	columns	container_width_rule	gutter	margin	notes
web	{{grid.web.columns}}	{{grid.web.container}}	{{grid.web.gutter}}	{{grid.web.margin}}	{{grid.web.notes}}
mobile	{{grid.mobile.columns}}	{{grid.mobile.container}}	{{grid.mobile.gutter}}	{{grid.mobile.margin}}	{{grid.mobile.notes}}

2) Spacing Token Usage Rules (required)

- Baseline unit: {{spacing.baseline_unit}} (references {{xref:DSYS-01}})
- Allowed spacing tokens: {{spacing.allowed_tokens}}
- When to use larger spacing: {{spacing.when_large}}
- When to use tighter spacing: {{spacing.when_tight}}

3) Density Modes (optional)

mode	description	intended_surfaces	constraints
comfortable	{{density.comfortable.desc}}	{{density.comfortable.surfaces}}	{{density.comfortable.constraints}}
compact	{{density.compact.desc}}	{{density.compact.surfaces}}	{{density.compact.constraints}}

4) Alignment Rules (required)

- Edge alignment rule: {{alignment.edges}}
- Baseline alignment rule: {{alignment.baseline}} | OPTIONAL
- Center alignment rule: {{alignment.center}} | OPTIONAL

5) Layout Patterns (required)

pattern	structure	spacing_guidance	do	dont
list	{{patterns.list.structure}}	{{patterns.list.spacing}}	{{patterns.list.do}}	{{patterns.list.dont}}
form	{{patterns.form.structure}}	{{patterns.form.spacing}}	{{patterns.form.do}}	{{patterns.form.dont}}
card	{{patterns.card.structure}}	{{patterns.card.spacing}}	{{patterns.card.do}}	{{patterns.card.dont}}

6) Touch Target Spacing (required)

- Minimum touch target size: {{touch.min_target}}
- Minimum spacing between targets: {{touch.min_spacing}}
- Dense UI rule: {{touch.dense_rule}} | OPTIONAL

7) Exceptions Policy (required)

- When exceptions are allowed: {{exceptions.when_allowed}}
- Required justification fields: {{exceptions.justification_fields}}
- Who approves exceptions: {{exceptions.approver}} | OPTIONAL

8) Anti-Patterns (optional)

- {{anti_patterns[0]}} | OPTIONAL
- {{anti_patterns[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DSYS-01}}, {{xref:RLB-01}} | OPTIONAL, {{xref:RLB-02}} | OPTIONAL
- Downstream: {{xref:DES-03}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:FE-06}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Grid basics + spacing token rules.
- intermediate: Required. Add layout patterns and touch spacing.
- advanced: Required. Add density modes, exceptions governance, and anti-patterns.

Unknown Handling

- UNKNOWN_ALLOWED: `density_modes`, `anti_patterns`, `platform_differences`, `approver`, `notes`
- If spacing token usage rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DSYS
- Pass conditions:
  - required_fields_present == true
  - grid_definition_present == true
  - spacing_token_rules_present == true
  - touch_spacing_present == true

- exceptions_policy_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DSYS-04

DSYS-04 — Iconography & Illustration Rules

Header Block

- template_id: DSYS-04
- title: Iconography & Illustration Rules
- type: design_system_tokens
- template_version: 1.0.0
- output_path: 10_app/design_system/DSYS-04_Iconography_Illustration_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.DSYS
- upstream_dependencies: ["DSYS-01", "CDX-01", "A11YD-01"]
- inputs_required: ["DSYS-01", "CDX-01", "A11YD-01", "VAP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the rules for icon and illustration usage so visuals are consistent, accessible, and semantically correct. This covers style constraints, usage guidance, labeling/accessibility, and how icons/illustrations map to meaning in the UI.

Inputs Required

- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- CDX-01: {{xref:CDX-01}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL

- VAP-01: {{xref:VAP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Icon style rules:
    - stroke/filled policy
    - weight consistency
    - corner/angle rules
    - grid size (e.g., 24px base) and scaling rules
- Icon semantic rules:
    - when icons can be used alone vs must include label
    - avoid ambiguous icons list
    - do/don’t examples
- Illustration rules:
    - allowed use cases (empty states, onboarding, marketing surfaces)
    - forbidden use cases (critical instructions, safety)
    - tone alignment with CDX-01
- Accessibility rules:
    - decorative vs informative classification
    - alt text rules
    - aria-label rules for icon-only buttons
- Asset naming and organization rules (ties to VAP)

Optional Fields

- Brand mascot/character rules | OPTIONAL
- Localization considerations for visuals | OPTIONAL
- Notes | OPTIONAL

Rules

- Icons must not be the only indicator of meaning in critical flows; use text where needed.
- If an icon is used as a button with no text, it must have an accessible label.
- Illustrations must not encode culture-specific meaning unless localized.
- Visual meaning must align with terminology and tone (CDX-01).

Output Format

1) Icon Style Rules (required)

- Base grid size: {{icons.grid_size}}
- Stroke policy: {{icons.stroke_policy}}
- Weight rules: {{icons.weight_rules}}
- Corner/angle rules: {{icons.corner_rules}} | OPTIONAL
- Scaling rules: {{icons.scaling_rules}}

2) Icon Semantics (required)

scenario	allowed_icon_only	required_label	rationale
navigation_tab	{{semantics.nav.icon_only}}	{{semantics.nav.label}}	{{semantics.nav.rationale}}

primary_cta	{{semantics.cta.icon_only}}	{{semantics.cta.label}}	{{semantics.cta.rationale}}
destructive_action	{{semantics.destructive.icon_only}}	{{semantics.destructive.label}}	{{semantics.destructive.rationale}}

3) Ambiguous / Avoid List (required)

- {{icons.avoid[0]}}
- {{icons.avoid[1]}}
- {{icons.avoid[2]}}
- {{icons.avoid[3]}} | OPTIONAL
- {{icons.avoid[4]}} | OPTIONAL

4) Do/Don’t Examples (required)

example_id	do	dont	why
ex_01	{{examples[0].do}}	{{examples[0].dont}}	{{examples[0].why}}
ex_02	{{examples[1].do}}	{{examples[1].dont}}	{{examples[1].why}}

5) Illustration Rules (required)

- Allowed use cases: {{illustrations.allowed_use_cases}}
- Forbidden use cases: {{illustrations.forbidden_use_cases}}
- Tone alignment rules: {{illustrations.tone_alignment}}
- Complexity constraints (keep simple): {{illustrations.complexity}} | OPTIONAL

6) Accessibility Rules (required)

- Decorative visuals: {{a11y.decorative_rules}}

- Informative visuals: {{a11y.informative_rules}}
- Icon-only controls labeling: {{a11y.icon_only_labeling}}
- Alt text guidelines: {{a11y.alt_text_rules}} | OPTIONAL

7) Naming / Organization (required)

- Naming convention: {{assets.naming_convention}}
- Folder organization: {{assets.organization}} | OPTIONAL
- Source of truth: {{xref:VAP-01}} | OPTIONAL

Cross-References

- Upstream: {{xref:DSYS-01}} | OPTIONAL, {{xref:CDX-01}} | OPTIONAL
- Downstream: {{xref:VAP-02}} | OPTIONAL, {{xref:DES-03}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define icon rules + avoid list + a11y labeling.
- intermediate: Required. Add semantics table and do/don't examples.
- advanced: Required. Add illustration constraints and naming/organization rules tied to asset pipeline.

Unknown Handling

- UNKNOWN_ALLOWED: `mascot_rules`, `visual_localization_notes`, `notes`, `organization`

- If icon-only labeling rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DSYS
- Pass conditions:
    - required_fields_present == true
    - icon_style_rules_present == true
    - semantics_rules_present == true
    - a11y_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DSYS-05

DSYS-05 — Theming Rules (light/dark, brand constraints)

Header Block

- template_id: DSYS-05
- title: Theming Rules (light/dark, brand constraints)
- type: design_system_tokens
- template_version: 1.0.0
- output_path: 10_app/design_system/DSYS-05_Theming_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.DSYS
- upstream_dependencies: ["DSYS-01", "A11YD-04", "CDX-01"]
- inputs_required: ["DSYS-01", "A11YD-04", "CDX-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the enforceable theming model (light/dark/brand variations) so UI can switch themes without breaking semantics, accessibility, or product identity. This document sets rules for token resolution, contrast, and brand constraints.

Inputs Required

- DSYS-01: {{xref:DSYS-01}}
- A11YD-04: {{xref:A11YD-04}} | OPTIONAL
- CDX-01: {{xref:CDX-01}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Theme list (minimum 1; typically light + dark)
- For each theme:
  - theme_id
  - intended environments (system/default/user choice)
  - token value source (DSYS-01 semantic tokens)
  - contrast compliance rules (text, icons, controls)
- Theme switching rules:
  - system preference handling
  - app override handling
  - persistence rules
- Brand constraints:
  - non-negotiable tokens (brand identity anchors)
  - allowed variation range (where theming can differ)
  - forbidden changes (e.g., status colors must remain semantic)
- Component theming rules:
  - which components can theme independently (if any)
  - state token handling across themes (hover/focus/disabled)
- Visual regression expectations (what must be tested)

Optional Fields

- Seasonal/campaign themes | OPTIONAL
- High-contrast theme support | OPTIONAL
- Notes | OPTIONAL

Rules

- Theme values must map to semantic tokens, not direct “raw” values in components.
- Contrast requirements apply in every theme; exceptions are not allowed for text/controls.
- Brand anchors must be preserved across themes (defined explicitly).
- If app supports manual theme selection, OS preference must still be respected unless explicitly overridden.

Output Format

1) Themes (required)

theme_id	name	default_source	user_selectable	persists	notes
theme_light	{{themes.light.name}}	{{themes.light.source}}	{{themes.light.user_selectable}}	{{themes.light.persists}}	{{themes.light.notes}}
theme_dark	{{themes.dark.name}}	{{themes.dark.source}}	{{themes.dark.user_selectable}}	{{themes.dark.persists}}	{{themes.dark.notes}}

2) Switching Rules (required)

- System preference behavior: {{switching.system_pref}}
- App override behavior: {{switching.app_override}}
- Persistence: {{switching.persistence}}
- Fallback behavior: {{switching.fallback}}

3) Brand Constraints (required)

- Brand anchors (must not change): {{brand.anchors}}
- Allowed variation: {{brand.allowed_variation}}
- Forbidden changes: {{brand.forbidden}}

4) Token Resolution Rules (required)

- Components consume semantic tokens only: {{resolution.semantic_only}}
- State tokens in each theme: {{resolution.state_tokens}}
- “Custom per component” policy: {{resolution.custom_policy}} | OPTIONAL

5) Contrast Compliance (required)

- Text contrast requirement: {{contrast.text}}
- Icon contrast requirement: {{contrast.icons}}
- Control contrast requirement: {{contrast.controls}}
- Verification method: {{contrast.verification_method}} | OPTIONAL

6) Component Theming Rules (required)

component_id	theming_allowed	theme_variant_rules	state_handling_notes
{{components[0].id}}	{{components[0].allowed}}	{{components[0].rules}}	{{components[0].state_notes}}

7) Visual Regression Expectations (required)

- Screens to snapshot: {{regression.screens}}
- Components to snapshot: {{regression.components}} | OPTIONAL

- Failure threshold policy: {{regression.threshold_policy}} | OPTIONAL

8) Optional Themes (optional)

- Seasonal/campaign themes: {{optional.seasonal}} | OPTIONAL
- High-contrast: {{optional.high_contrast}} | OPTIONAL

Cross-References

- Upstream: {{xref:DSYS-01}}, {{xref:A11YD-04}} | OPTIONAL
- Downstream: {{xref:FE-06}} | OPTIONAL, {{xref:MOB-*}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define themes + switching rules + brand anchors.
- intermediate: Required. Add token resolution and contrast rules.
- advanced: Required. Add component theming allowances and regression expectations.

Unknown Handling

- UNKNOWN_ALLOWED: `seasonal_themes`, `high_contrast`, `verification_method`, `threshold_policy`, `notes`
- If brand anchors or contrast rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DSYS
- Pass conditions:
    - required_fields_present == true
    - themes_defined == true
    - switching_rules_present == true
    - brand_constraints_present == true
    - contrast_rules_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

Information Architecture & Navigation (IAN)

Information Architecture & Navigation (IAN)

- IAN-01 Navigation Map (primary/secondary, tabs/drawers)
- IAN-02 Route & Deep Link Spec (route IDs, params)
- IAN-03 Information Architecture Tree (sections/pages hierarchy)
- IAN-04 Search/Filter/Sort UX (if applicable)
- IAN-05 Access-Gated Navigation Rules (role-based visibility)

IAN-01

IAN-01 — Navigation Map (primary/secondary, tabs/drawers)

Header Block

- template_id: IAN-01
- title: Navigation Map (primary/secondary, tabs/drawers)
- type: information_architecture_navigation
- template_version: 1.0.0
- output_path: 10_app/ia/IAN-01_Navigation_Map.md
- compliance_gate_id: TMP-05.PRIMARY.IAN
- upstream_dependencies: ["PRD-04", "DES-01", "PRD-03"]
- inputs_required: ["PRD-04", "DES-01", "PRD-03", "RSC-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical navigation structure users experience: primary navigation surfaces, secondary navigation, and the rules that govern how users move between sections. This is the source of truth for navigation intent and structure (not implementation code).

Inputs Required

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- DES-01: {{xref:DES-01}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL

- RSC-02: {{xref:RSC-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Navigation paradigm(s) used (tabs, drawer, stack, sidebar, top nav)
- Primary navigation items (minimum 3)
- Secondary navigation items (if any)
- For each nav item:
    - nav_id
    - label (ties to CDX inventory later)
    - destination (screen_id or route_id)
    - visibility rules (role/tier/access)
    - ordering/priority
    - icon (optional pointer)
    - badges/indicators rules (optional)
- Global navigation rules:
    - default landing
    - back behavior policy (high level)
    - how modal overlays interact with nav
    - auth gating behavior (what happens when not signed in)
- Role-based navigation differences (if applicable)

Optional Fields

- Search entry points | OPTIONAL
- Deep link entry points | OPTIONAL
- Notes | OPTIONAL

Rules

- Do not invent roles; use PRD-03 / IAM concepts.
- Destinations should use screen_id where possible; route_id is finalized in IAN-02.
- Navigation must avoid dead ends: every destination must offer an exit path.
- If visibility rules exist, define a deterministic fallback (hide vs disabled vs upsell).

Output Format

1) Navigation Overview (required)

- Paradigms: {{nav.paradigms}}
- Default landing destination: {{nav.default_landing}} (screen_id or route_id)
- Auth gating behavior: {{nav.auth_gating_behavior}}

2) Primary Navigation (required)

nav_id	label	destination_screen_id	destination_route_id	visibility_rules	order	icon_ref	badge_rules	notes
nav_01	{{primary[0].label}}	{{primary[0].screen_id}}	{{primary[0].route_id}}	{{primary[0].visibility}}	{{primary[0].order}}	{{primary[0].icon}}	{{primary[0].badge}}	{{primary[0].notes}}

nav_02	{{primary[1].label}}	{{primary[1].screen_id}}	{{primary[1].route_id}}	{{primary[1].visibility}}	{{primary[1].order}}	{{primary[1].icon}}	{{primary[1].badge}}	{{primary[1].notes}}

3) Secondary Navigation (optional)

nav_id	label	destination_screen_id	destination_route_id	visibility_rules	order	notes
nav_s01	{{secondary[0].label}}	{{secondary[0].screen_id}}	{{secondary[0].route_id}}	{{secondary[0].visibility}}	{{secondary[0].order}}	{{secondary[0].notes}}

4) Global Navigation Rules (required)

- Back behavior policy: {{rules.back_behavior}}
- Modal overlay policy: {{rules.modal_policy}}
- Dead-end prevention rule: {{rules.no_dead_ends}}
- Fallback when not visible: {{rules.visibility_fallback}} (hide/disable/upsell)

5) Role/Tier Variants (required if applicable)

variant_id	applies_to	differences	notes
var_01	{{variants[0].applies_to}}	{{variants[0].diff}}	{{variants[0].notes}}

Cross-References

- Upstream: {{xref:PRD-03}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL, {{xref:DES-01}} | OPTIONAL, {{xref:RSC-02}} | OPTIONAL
- Downstream: {{xref:IAN-02}}, {{xref:DES-02}} | OPTIONAL, {{xref:MAP-01}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:MOB-01}} | OPTIONAL

- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Primary nav + default landing + auth gating.
- intermediate: Required. Add visibility rules and global policies.
- advanced: Required. Add variants, badge rules, and deeper policy definitions.

Unknown Handling

- UNKNOWN_ALLOWED: `secondary_nav`, `icon_ref`, `badge_rules`, `variants`, `notes`, `route_id` (until IAN-02)
- If default landing is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IAN
- Pass conditions:
    - required_fields_present == true
    - primary_nav_count >= 3
    - default_landing_present == true
    - no_dead_ends_rule_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

IAN-02

IAN-02 — Route & Deep Link Spec (route IDs, params)

Header Block

- template_id: IAN-02
- title: Route & Deep Link Spec (route IDs, params)
- type: information_architecture_navigation
- template_version: 1.0.0
- output_path: 10_app/ia/IAN-02_Route_DeepLink_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.IAN
- upstream_dependencies: ["IAN-01", "DES-02"]
- inputs_required: ["IAN-01", "DES-02", "PRD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical routing contract: route IDs/paths, parameters, and deep link behavior. This makes navigation implementable and prevents route drift across web/mobile and across releases.

Inputs Required

- IAN-01: {{xref:IAN-01}}
- DES-02: {{xref:DES-02}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Route naming convention (IDs + optional path patterns)
- Route list (minimum equals DES-02 screens unless explicitly “screenless”)
- For each route:
    - route_id
    - screen_id
    - path pattern (web) | OPTIONAL
    - deep link pattern(s) (mobile) | OPTIONAL
    - params schema (name/type/required/default)
    - access requirements (role/entitlement)
    - navigation type (push/replace/modal/tab)
    - canonical back target (where “back” goes)
    - analytics hook (screen_view event name) | OPTIONAL
- Deep link behavior rules:
    - cold start handling
    - auth gating handling
    - invalid param handling
    - unknown route handling

Optional Fields

- Legacy route aliases | OPTIONAL
- SEO notes (web) | OPTIONAL
- Notes | OPTIONAL

Rules

- route_id must be stable and never reused for a different screen.
- Params must be explicitly typed; avoid “stringly typed” ambiguity.
- Deep links must have deterministic fallbacks (e.g., send to safe landing).
- Access requirements must align with PRD-03/IAM/BRP entitlements; do not invent.
- If a route is removed, define deprecation/redirect policy (REL).

Output Format

1) Naming Conventions (required)

- Route ID format: {{routes.naming.id_format}} (e.g., route_<slug>)
- Path format (web): {{routes.naming.web_path_format}} | OPTIONAL
- Deep link scheme/host: {{routes.naming.deeplink_scheme}} | OPTIONAL

2) Route Catalog (canonical)

route_id	screen_id	path_pattern	deeplink_patterns	params_schema	nav_type	access_req	back_target	analytics_event	notes
route_	{{routes[0].screen_id}}	{{routes[0].path}}	{{routes[0].deeplinks}}	{{routes[0].params}}	{{routes[0].nav_type}}	{{routes[0].access}}	{{routes[0].back_target}}	{{routes[0].analytics}}	{{routes[0].notes}}

route_01	{{routes[1].screen_id}}	{{routes[1].path}}	{{routes[1].deeplinks}}	{{routes[1].params}}	{{routes[1].nav_type}}	{{routes[1].access}}	{{routes[1].back_target}}	{{routes[1].analytics}}	{{routes[1].notes}}
route_02									

3) Params Schema Detail (required)

For complex params, define explicit schema blocks.

{{routes[0].route_id}} params

param	type	required	default	validation	notes
{{routes[0].params_detail[0].name}}	{{routes[0].params_detail[0].type}}	{{routes[0].params_detail[0].required}}	{{routes[0].params_detail[0].default}}	{{routes[0].params_detail[0].validation}}	{{routes[0].params_detail[0].notes}}

4) Deep Link Rules (required)

- Cold start routing: {{deeplink.cold_start}}
- Auth gating behavior: {{deeplink.auth_gating}}
- Invalid params: {{deeplink.invalid_params}}
- Unknown route: {{deeplink.unknown_route}}
- Post-auth continuation rule: {{deeplink.post_auth_continue}} | OPTIONAL

5) Deprecation / Redirect Rules (optional)

- Legacy aliases: {{legacy.aliases}} | OPTIONAL
- Redirect policy: {{legacy.redirect_policy}} | OPTIONAL

Cross-References

- Upstream: {{xref:IAN-01}}, {{xref:DES-02}} | OPTIONAL
- Downstream: {{xref:FE-01}} | OPTIONAL, {{xref:MOB-01}} | OPTIONAL, {{xref:ROUTE-*}} | OPTIONAL, {{xref:REL-02}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Route catalog with IDs, screens, and basic params.
- intermediate: Required. Add deep link rules and access requirements.
- advanced: Required. Add deprecation rules, analytics hooks, and typed schema details.

Unknown Handling

- UNKNOWN_ALLOWED: `path_pattern`, `deeplink_patterns`, `analytics_event`, `legacy_aliases`, `seo_notes`, `notes`
- If any route lacks route_id or screen_id mapping → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IAN
- Pass conditions:
    - required_fields_present == true
    - route_ids_unique == true
    - every_screen_has_route == true
    - deeplink_rules_present == true
    - placeholder_resolution == true

- no_unapproved_unknowns == true

IAN-03

IAN-03 — Information Architecture Tree (sections/pages hierarchy)

Header Block

- template_id: IAN-03
- title: Information Architecture Tree (sections/pages hierarchy)
- type: information_architecture_navigation
- template_version: 1.0.0
- output_path: 10_app/ia/IAN-03_IA_Tree.md
- compliance_gate_id: TMP-05.PRIMARY.IAN
- upstream_dependencies: ["IAN-01", "DES-02", "URD-03"]
- inputs_required: ["IAN-01", "DES-02", "URD-03", "PRD-04", "CDX-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical information hierarchy: top-level sections, sub-sections, and the screens/pages that live within each. This makes the product navigable, supports labeling consistency, and prevents duplicated or orphaned surfaces.

Inputs Required

- IAN-01: {{xref:IAN-01}} | OPTIONAL
- DES-02: {{xref:DES-02}} | OPTIONAL
- URD-03: {{xref:URD-03}} | OPTIONAL

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- CDX-01: {{xref:CDX-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- IA tree (minimum 2 levels deep for non-trivial products)
- Each node includes:
    - node_id
    - label (or label key)
    - type (section/subsection/page/group)
    - destination (screen_id/route_id) if navigable
    - visibility rules (role/tier/access)
    - ordering
- Cross-links (where a screen appears in multiple contexts) with primary home
- Coverage check: every screen_id from DES-02 appears in exactly one primary place (can have secondary links)

Optional Fields

- Search taxonomy (tags/categories) | OPTIONAL
- Notes | OPTIONAL

Rules

- IA labels must align with CDX-01 style rules; final strings live in CDX-02.

- Avoid duplicating screens in multiple primary homes; if needed, choose one primary and list others as secondary links.
- Visibility rules must be deterministic (no “depends”).
- If a screen is “utility” (settings, help), it must still have a home location.

Output Format

1) IA Tree (required)

Use a structured outline plus a canonical table.

Outline

- {{ia.root.label}}
    - {{ia.root.children[0].label}}
        - {{ia.root.children[0].children[0].label}} (screen: {{ia.root.children[0].children[0].screen_id}}) | OPTIONAL
    - {{ia.root.children[1].label}} | OPTIONAL

2) IA Nodes Table (canonical)

node_id	label	type	destination_screen_id	destination_route_id	visibility_rules	order	primary_home	notes
node_01	{{nodes[0].label}}	{{nodes[0].type}}	{{nodes[0].screen_id}}	{{nodes[0].route_id}}	{{nodes[0].visibility}}	{{nodes[0].order}}	{{nodes[0].primary_home}}	{{nodes[0].notes}}
node_02	{{nodes[1].label}}	{{nodes[1].type}}	{{nodes[1].screen_id}}	{{nodes[1].route_id}}	{{nodes[1].visibility}}	{{nodes[1].order}}	{{nodes[1].primary_home}}	{{nodes[1].notes}}

3) Cross-links (required if any)

screen_id	primary_node_id	secondary_node_ids	rationale
{{crosslinks[0].screen_id}}	{{crosslinks[0].primary}}	{{crosslinks[0].secondary}}	{{crosslinks[0].rationale}}

4) Coverage Checks (required)

- Every DES-02 screen has a primary node: {{coverage.all_screens_have_primary}}
- No screen has multiple primaries: {{coverage.no_multiple_primary}}
- Utility screens placed: {{coverage.utility_placed}} | OPTIONAL

5) Search Taxonomy (optional)

- Tags/categories: {{search.taxonomy}} | OPTIONAL

Cross-References

- Upstream: {{xref:IAN-01}}, {{xref:DES-02}} | OPTIONAL, {{xref:URD-03}} | OPTIONAL
- Downstream: {{xref:IAN-02}} | OPTIONAL, {{xref:CDX-02}} | OPTIONAL, {{xref:DISC-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Basic tree + node table for primary sections and pages.
- intermediate: Required. Add visibility rules and cross-links.
- advanced: Required. Add coverage checks and search taxonomy if applicable.

Unknown Handling

- UNKNOWN_ALLOWED: `search_taxonomy`, `notes`, `secondary_links`, `destination_route_id` (until IAN-02)
- If any DES-02 screen lacks a primary node → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IAN
- Pass conditions:
    - required_fields_present == true
    - tree_present == true
    - node_ids_unique == true
    - all_screens_have_primary == true
    - no_multiple_primary == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

IAN-04

IAN-04 — Search/Filter/Sort UX (if applicable)

Header Block

- template_id: IAN-04
- title: Search/Filter/Sort UX (if applicable)
- type: information_architecture_navigation
- template_version: 1.0.0
- output_path: 10_app/ia/IAN-04_Search_Filter_Sort_UX.md
- compliance_gate_id: TMP-05.PRIMARY.IAN
- upstream_dependencies: ["IAN-03", "DES-03", "URD-03"]
- inputs_required: ["IAN-03", "DES-03", "URD-03", "PRD-04", "CDX-01", "CDX-02", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define the user experience contract for search, filtering, and sorting across the product (when applicable): where search exists, what it searches, how filters/sorts behave, and what users see in empty/error states. This is UX-authoritative and feeds implementation and search/index design.

Inputs Required

- IAN-03: {{xref:IAN-03}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL

- URD-03: {{xref:URD-03}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- CDX-01: {{xref:CDX-01}} | OPTIONAL
- CDX-02: {{xref:CDX-02}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false). If false, explicitly mark N/A.
- Search surfaces list (screens/components where search exists)
- For each search surface:
    - surface_id (screen_id/component_id)
    - search scope (what entities/content)
    - query input rules (min chars, debounce, submit behavior)
    - results presentation (list/cards/groups)
    - ranking expectation (basic relevance / recency / custom)
    - empty states (no results, no data)
    - error states (network/server)
    - accessibility behavior (keyboard nav, SR announcements)
- Filter catalog (if applicable):
    - filter_id, label key, type (multi/single/range), defaults, persistence rules
- Sort catalog (if applicable):

    - sort_id, label key, options, default, persistence rules
- Persistence rules (remember last query/filter/sort or not)
- Reset/clear behavior

Optional Fields

- Advanced search operators | OPTIONAL
- Saved searches | OPTIONAL
- Notes | OPTIONAL

Rules

- If applies == false, include 00_NA block and do not define catalogs.
- Search and filter labels must follow CDX rules; final strings in CDX-02.
- Must specify deterministic behavior for: debounce, submit, clear, back behavior.
- Must include accessibility rules for keyboard-only use.
- Empty/error state UX must align with DES-05 and CDX catalogs.

Output Format

1) Applicability

- applies: {{search.applies}} (true/false)
- rationale: {{search.rationale}} | OPTIONAL
- 00_NA (if not applies): {{search.na_block}} | OPTIONAL

2) Search Surfaces (if applies)

surface_id	surface_type	scope	input_rules	results_presentation	ranking_expectation	empty_states	error_states	a11y_behavior	notes
{{surfaces[0].id}}	{{surfaces[0].type}}	{{surfaces[0].scope}}	{{surfaces[0].input}}	{{surfaces[0].results}}	{{surfaces[0].ranking}}	{{surfaces[0].empty}}	{{surfaces[0].errors}}	{{surfaces[0].a11y}}	{{surfaces[0].notes}}

3) Filters Catalog (if applies)

filter_id	label_key	type	options_or_range	default	persistence	clear_behavior	notes
{{filters[0].id}}	{{filters[0].label_key}}	{{filters[0].type}}	{{filters[0].options}}	{{filters[0].default}}	{{filters[0].persistence}}	{{filters[0].clear}}	{{filters[0].notes}}

4) Sort Catalog (if applies)

sort_id	label_key	options	default	persistence	notes
{{sorts[0].id}}	{{sorts[0].label_key}}	{{sorts[0].options}}	{{sorts[0].default}}	{{sorts[0].persistence}}	{{sorts[0].notes}}

5) Persistence + Reset Rules (if applies)

- Remember last query: {{rules.remember_query}}
- Remember filters: {{rules.remember_filters}}
- Remember sort: {{rules.remember_sort}}
- Clear all behavior: {{rules.clear_all}}
- Back navigation behavior: {{rules.back_behavior}} | OPTIONAL

6) Accessibility Requirements (required if applies)

- Keyboard navigation: {{a11y.keyboard}}
- Screen reader announcements: {{a11y.sr}}
- Focus management for results updates: {{a11y.focus}} | OPTIONAL

Cross-References

- Upstream: {{xref:IAN-03}} | OPTIONAL, {{xref:DES-05}} | OPTIONAL, {{xref:CDX-02}} | OPTIONAL

- Downstream: {{xref:DISC-*}}* | *OPTIONAL, {{xref:SRCH-}}* | OPTIONAL, {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL

- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required unless search exists.

- intermediate: Required if applies. Define surfaces + basic filters/sorts.

- advanced: Required if applies. Add determinism for ranking expectations and persistence rules.

Unknown Handling

- UNKNOWN_ALLOWED: `advanced_operators`, `saved_searches`, `notes`, `ranking_expectation` (if basic relevance), `persistence_rules` (must still be stated)

- If applies == true and input_rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IAN

- Pass conditions:

    - required_fields_present == true

    - if_applies_then_surfaces_present == true

- if_applies_then_a11y_defined == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

IAN-5

IAN-05 — Access-Gated Navigation Rules (role-based visibility)

Header Block

- template_id: IAN-05
- title: Access-Gated Navigation Rules (role-based visibility)
- type: information_architecture_navigation
- template_version: 1.0.0
- output_path: 10_app/ia/IAN-05_Access_Gated_Navigation_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.IAN
- upstream_dependencies: ["IAN-01", "PRD-03", "BRP-02", "IAM-03"]
- inputs_required: ["IAN-01", "PRD-03", "BRP-02", "IAM-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic rules for how navigation behaves when access is restricted: what is hidden vs disabled, how upsells or explanations are shown, and what happens if a user deep-links into restricted content. This prevents inconsistent access handling across the product.

Inputs Required

- IAN-01: {{xref:IAN-01}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- BRP-02: {{xref:BRP-02}} | OPTIONAL

- IAM-03: {{xref:IAM-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Access gating display modes:
    - hide
    - disable (show but inactive)
    - show-with-upsell (if monetized)
    - show-with-request-access (if applicable)
- Deterministic rule table for which mode applies by nav surface type:
    - primary nav
    - secondary nav
    - contextual links
    - deep links
- Deep link restricted behavior:
    - redirect target
    - explanation UI
    - logging/telemetry requirement
- Copy requirements pointer (CDX) for access messages
- Security requirement: avoid leaking details of restricted resources

Optional Fields

- Org-level policies (enterprise access) | OPTIONAL
- Audit trail needs | OPTIONAL
- Notes | OPTIONAL

Rules

- Must not invent roles or entitlements; use PRD-03/BRP-02/IAM.
- Deep links must be safe by default: no partial rendering of restricted data.
- If showing disabled items, there must be an accessible explanation (tooltip/help text) and a deterministic action if any (upgrade/request).
- If upsell exists, it must be consistent with pricing policy (BRP-03 / REVOPS).

Output Format

1) Gating Modes (required)

mode	description	user_experience	when_used	notes
hide	{{modes.hide.desc}}	{{modes.hide.ux}}	{{modes.hide.when}}	{{modes.hide.notes}}
disable	{{modes.disable.desc}}	{{modes.disable.ux}}	{{modes.disable.when}}	{{modes.disable.notes}}
upsell	{{modes.upsell.desc}}	{{modes.upsell.ux}}	{{modes.upsell.when}}	{{modes.upsell.notes}}
request_access	{{modes.request.desc}}	{{modes.request.ux}}	{{modes.request.when}}	{{modes.request.notes}}

2) Rule Table by Surface (required)

surface_type	default_mode	overrides	rationale

primary_nav	{{rules.primary.default_mode}}	{{rules.primary.overrides}}	{{rules.primary.rationale}}
secondary_nav	{{rules.secondary.default_mode}}	{{rules.secondary.overrides}}	{{rules.secondary.rationale}}
contextual_link	{{rules.contextual.default_mode}}	{{rules.contextual.overrides}}	{{rules.contextual.rationale}}
deep_link	{{rules.deeplink.default_mode}}	{{rules.deeplink.overrides}}	{{rules.deeplink.rationale}}

3) Deep Link Restricted Handling (required)

- Redirect target: {{deeplink.redirect_target}} (route_id/screen_id)
- Explanation UI: {{deeplink.explanation_ui}} (banner/screen/modal)
- User action offered: {{deeplink.user_action}} (login/upgrade/request access/back)
- Telemetry/logging: {{deeplink.telemetry}}
- Security note (no leakage): {{deeplink.no_leakage_rule}}

4) Copy Requirements (required)

- Access denied message source: {{xref:CDX-04}} | OPTIONAL
- Upsell/request-access message source: {{xref:CDX-02}} | OPTIONAL

5) Compliance Checks (required)

- All restricted destinations have deterministic handling: {{checks.coverage_complete}}
- No restricted resource data shown pre-check: {{checks.no_precheck_render}}

Cross-References

- Upstream: {{xref:IAN-01}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL, {{xref:BRP-02}} | OPTIONAL, {{xref:IAM-03}} | OPTIONAL

- Downstream: {{xref:IAN-02}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:MOB-01}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL

- Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define modes + deep link behavior + default rules.

- intermediate: Required. Add overrides and copy pointers.

- advanced: Required. Add telemetry + security checks and enterprise policy notes if needed.

Unknown Handling

- UNKNOWN_ALLOWED: `org_policies`, `audit_trail_needs`, `notes`, `overrides`

- If deep link restricted handling is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.IAN

- Pass conditions:

    - required_fields_present == true

    - gating_modes_present == true

    - rule_table_present == true

    - deeplink_handling_present == true

    - coverage_complete == true

    - placeholder_resolution == true

- no_unapproved_unknowns == true

Accessibility Design (A11YD)

Accessibility Design (A11YD)

- A11YD-01 Accessibility Requirements Checklist (WCAG-aligned)
- A11YD-02 Keyboard/Focus Order Spec
- A11YD-03 Screen Reader & Labels Spec (aria/alt/roles)
- A11YD-04 Color Contrast & Visual Accessibility Spec
- A11YD-05 Accessible Error Messaging Rules

A11YD-01

A11YD-01 — Accessibility Requirements Checklist (WCAG-aligned)

Header Block

- template_id: A11YD-01
- title: Accessibility Requirements Checklist (WCAG-aligned)
- type: accessibility_design
- template_version: 1.0.0
- output_path: 10_app/accessibility/A11YD-01_A11y_Requirements_Checklist.md
- compliance_gate_id: TMP-05.PRIMARY.A11Y
- upstream_dependencies: ["PRD-06", "DES-03", "DSYS-01"]
- inputs_required: ["PRD-06", "DES-03", "DSYS-01", "CDX-01", "IAN-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the product's accessibility requirements as a testable checklist aligned to WCAG principles. This is the canonical accessibility contract that informs design, implementation, and QA gates.

Inputs Required

- PRD-06: {{xref:PRD-06}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL

- CDX-01: {{xref:CDX-01}} | OPTIONAL
- IAN-01: {{xref:IAN-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Target compliance level (e.g., WCAG 2.1 AA) (or product-specific)
- Checklist categories (minimum):
    - perceivable (text alternatives, contrast)
    - operable (keyboard, focus, timing)
    - understandable (labels, errors)
    - robust (semantics, compatibility)
- For each requirement:
    - req_id
    - category
    - requirement statement (testable)
    - applies_to (screens/components/global)
    - verification method (manual/automated)
    - severity (must/should)
    - owner (design/dev/qa)
    - references (screen_id/component_id or doc pointers)
- Known exceptions (must be justified)

Optional Fields

- Assistive tech support targets (SR/browser/device list) | OPTIONAL
- Training notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Requirements must be testable; avoid vague “be accessible” language.
- Exceptions must be explicitly documented and linked to decision/approval (STK-02/04).
- Accessibility requirements override motion/visual preferences when conflicting.
- Must include reduced motion requirement (links to IXD-05).

Output Format

1) Compliance Target (required)

- Target standard: {{a11y.target_standard}}
- Level: {{a11y.level}}
- Scope statement: {{a11y.scope_statement}} | OPTIONAL

2) Checklist (canonical)

req_id	category	requirement	applies_to	verification	severity	owner	refs	notes
a11y_req_001	perceivable	{{reqs[0].text}}	{{reqs[0].applies_to}}	{{reqs[0].verify}}	{{reqs[0].severity}}	{{reqs[0].owner}}	{{reqs[0].refs}}	{{reqs[0].notes}}
a11y_req_002	operable	{{reqs[1].text}}	{{reqs[1].applies_to}}	{{reqs[1].verify}}	{{reqs[1].severity}}	{{reqs[1].owner}}	{{reqs[1].refs}}	{{reqs[1].notes}}

3) Assistive Tech Targets (optional)

- Screen readers: {{targets.screen_readers}} | OPTIONAL
- Browsers: {{targets.browsers}} | OPTIONAL
- Mobile OS versions: {{targets.mobile_os}} | OPTIONAL
- Input devices: {{targets.input_devices}} | OPTIONAL

4) Exceptions (required if any)

exception_id	requirement_req_id	scope	justification	approval_ref
ex_01	{{exceptions[0].req_id}}	{{exceptions[0].scope}}	{{exceptions[0].why}}	{{exceptions[0].approval}}

5) Coverage Checks (required)

- Reduced motion covered (IXD-05): {{coverage.reduce_motion}}
- Keyboard/focus covered: {{coverage.keyboard_focus}}
- Contrast covered (A11YD-04): {{coverage.contrast}}
- Error messaging covered (A11YD-05): {{coverage.errors}}

Cross-References

- Upstream: {{xref:PRD-06}} | OPTIONAL, {{xref:DSYS-01}} | OPTIONAL, {{xref:DES-03}} | OPTIONAL
- Downstream: {{xref:A11YD-02}}, {{xref:A11YD-03}}, {{xref:A11YD-04}}, {{xref:A11YD-05}}, {{xref:IXD-05}} | OPTIONAL, {{xref:QA-*}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Checklist with must/should + verification.
- intermediate: Required. Add assistive tech targets and refs to screens/components.
- advanced: Required. Add exception governance and tighter verification methods.

Unknown Handling

- UNKNOWN_ALLOWED: `assistive_tech_targets`, `training_notes`, `notes`, `exceptions` (if none, explicitly state none)
- If target standard/level is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.A11Y
- Pass conditions:
    - required_fields_present == true
    - compliance_target_present == true
    - checklist_present == true
    - requirements_testable == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

A11YD-02

A11YD-02 — Keyboard/Focus Order Spec

Header Block

- template_id: A11YD-02
- title: Keyboard/Focus Order Spec
- type: accessibility_design
- template_version: 1.0.0
- output_path: 10_app/accessibility/A11YD-02_Keyboard_Focus_Order_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.A11Y
- upstream_dependencies: ["DES-02", "DES-03", "IXD-01", "IAN-02"]
- inputs_required: ["DES-02", "DES-03", "IXD-01", "IAN-02", "DSYS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the keyboard navigation and focus behavior contract across screens and interaction patterns: tab order, focus traps, skip links, focus restoration, and how focus changes during state transitions.

Inputs Required

- DES-02: {{xref:DES-02}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL
- IXD-01: {{xref:IXD-01}} | OPTIONAL
- IAN-02: {{xref:IAN-02}} | OPTIONAL

- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Global keyboard rules:
    - tab order principles (logical reading order)
    - focus visibility rules (focus ring usage)
    - skip navigation behavior (web)
    - focus restoration rules (closing modals, route changes)
- Pattern-level focus rules:
    - modals (focus trap + escape)
    - drawers
    - menus
    - toasts/alerts
    - forms (error focus behavior)
- Screen-specific exceptions list (only if needed)
- Verification checklist

Optional Fields

- Shortcut keys (if supported) | OPTIONAL
- Notes | OPTIONAL

Rules

- Focus must never be lost; after any action, a focus target must exist.
- Keyboard access must provide functional equivalence to pointer/gesture.
- Opening an overlay must move focus into it; closing must restore focus to trigger or logical next.
- Error validation must move focus to first error or error summary per DES-06.
- If shortcuts exist, they must be discoverable and not conflict with assistive tech.

Output Format

1) Global Keyboard Rules (required)

- Tab order principle: {{keyboard.tab_order_principle}}
- Focus visible requirement: {{keyboard.focus_visible}}
- Skip link behavior (web): {{keyboard.skip_link}} | OPTIONAL
- Focus restoration: {{keyboard.focus_restore}}

2) Pattern Focus Rules (required)

pattern_id	focus_entry	focus_trap	escape_behavior	focus_restore	notes
pat_modal	{{patterns.modal.entry}}	{{patterns.modal.trap}}	{{patterns.modal.escape}}	{{patterns.modal.restore}}	{{patterns.modal.notes}}
pat_drawer	{{patterns.drawer.entry}}	{{patterns.drawer.trap}}	{{patterns.drawer.escape}}	{{patterns.drawer.restore}}	{{patterns.drawer.notes}}
pat_menu	{{patterns.menu.entry}}	{{patterns.menu.trap}}	{{patterns.menu.escape}}	{{patterns.menu.restore}}	{{patterns.menu.notes}}

3) Forms Focus Rules (required)

- Focus first error rule: {{forms.focus_first_error}}

- Error summary focus rule: {{forms.error_summary_focus}} | OPTIONAL
- Inline error announcement: {{forms.inline_announce}} | OPTIONAL

4) Screen Exceptions (optional)

screen_id	exception	rationale	mitigation
{{exceptions[0].screen_id}}	{{exceptions[0].exception}}	{{exceptions[0].rationale}}	{{exceptions[0].mitigation}}

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:IXD-01}} | OPTIONAL, {{xref:DES-06}} | OPTIONAL, {{xref:IAN-02}} | OPTIONAL
- Downstream: {{xref:A11YD-03}} | OPTIONAL, {{xref:FE-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global rules + modal/menu focus behavior + verification list.
- intermediate: Required. Add form error focus rules and restoration details.
- advanced: Required. Add exceptions handling and shortcut key policy (if used).

Unknown Handling

- UNKNOWN_ALLOWED: `skip_link`, `shortcuts`, `notes`, `exceptions`
- If focus restoration is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.A11Y
- Pass conditions:
    - required_fields_present == true
    - global_rules_present == true
    - pattern_rules_present == true
    - verification_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

A11YD-03

A11YD-03 — Screen Reader & Labels Spec (aria/alt/roles)

Header Block

- template_id: A11YD-03
- title: Screen Reader & Labels Spec (aria/alt/roles)
- type: accessibility_design
- template_version: 1.0.0
- output_path: 10_app/accessibility/A11YD-03_ScreenReader_Labels_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.A11Y
- upstream_dependencies: ["DES-03", "DSYS-02", "CDX-02"]
- inputs_required: ["DES-03", "DSYS-02", "CDX-02", "CDX-04", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the labeling and semantic requirements so screen readers and assistive tech can correctly interpret UI structure and controls. This includes aria/role expectations (web), accessibility labels (mobile), alt text rules, and announcement patterns.

Inputs Required

- DES-03: {{xref:DES-03}} | OPTIONAL
- DSYS-02: {{xref:DSYS-02}} | OPTIONAL
- CDX-02: {{xref:CDX-02}} | OPTIONAL

- CDX-04: {{xref:CDX-04}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Global labeling rules:
    - every interactive control has a programmatic name
    - icon-only control labeling
    - form label association rules
    - required/invalid state announcement rules
- Semantic structure rules:
    - headings hierarchy policy
    - landmark/region policy (web)
    - list/table semantics policy
- Component label contract (minimum 10 key component types)
- Alt text rules for images/illustrations/icons (decorative vs informative)
- Announcement rules (live regions / alert semantics) for:
    - errors
    - success confirmations
    - loading completion (if needed)
- Verification checklist

Optional Fields

- Content reading order rules | OPTIONAL
- Notes | OPTIONAL

Rules

- Labels must match CDX terminology; avoid labels that contradict visible text.
- Do not over-announce; announce changes only when user needs it.
- If a visible label is insufficient context, provide expanded accessible label.
- Decorative icons/images must not clutter SR output.

Output Format

1) Global Labeling Rules (required)

- Programmatic name required: {{labels.programmatic_name_required}}
- Icon-only labeling: {{labels.icon_only}}
- Form label association: {{labels.form_association}}
- Required/invalid announcement: {{labels.required_invalid_announce}}

2) Semantic Structure Rules (required)

- Headings hierarchy policy: {{semantics.headings}}
- Landmarks/regions (web): {{semantics.landmarks}} | OPTIONAL
- List semantics: {{semantics.lists}}
- Table semantics: {{semantics.tables}} | OPTIONAL

3) Component Label Contract (required)

component_type	required_label_source	aria_role_or_semantic	states_to_announce	example
button	{{components.button.label_source}}	{{components.button.role}}	{{components.button.states}}	{{components.button.example}}
text_input	{{components.input.label_source}}	{{components.input.role}}	{{components.input.states}}	{{components.input.example}}
checkbox	{{components.checkbox.label_source}}	{{components.checkbox.role}}	{{components.checkbox.states}}	{{components.checkbox.example}}
modal	{{components.modal.label_source}}	{{components.modal.role}}	{{components.modal.states}}	{{components.modal.example}}
toast/alert	{{components.alert.label_source}}	{{components.alert.role}}	{{components.alert.states}}	{{components.alert.example}}

4) Alt Text Rules (required)

- Decorative images/icons: {{alt.decorative}}
- Informative images/icons: {{alt.informative}}
- Illustrations in empty states: {{alt.empty_state_illustrations}} | OPTIONAL

5) Announcement Rules (required)

event_type	when_to_announce	mechanism	content_rule	example
error	{{announce.error.when}}	{{announce.error.mechanism}}	{{announce.error.content_rule}}	{{announce.error.example}}
success	{{announce.success.when}}	{{announce.success.mechanism}}	{{announce.success.content_rule}}	{{announce.success.example}}
status_change	{{announce.status.when}}	{{announce.status.mechanism}}	{{announce.status.content_rule}}	{{announce.status.example}}

6) Verification Checklist (required)

- {{verify[0]}}

- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:CDX-02}} | OPTIONAL, {{xref:CDX-04}} | OPTIONAL, {{xref:DSYS-02}} | OPTIONAL
- Downstream: {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Global labeling + alt rules + basic component contract.
- intermediate: Required. Add announcements and semantic structure rules.
- advanced: Required. Tighten examples and verification checklist to specific components/screens.

Unknown Handling

- UNKNOWN_ALLOWED: `landmarks`, `tables`, `reading_order`, `notes`
- If icon-only labeling rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.A11Y

- Pass conditions:
    - required_fields_present == true
    - component_contract_present == true
    - alt_text_rules_present == true
    - announcement_rules_present == true
    - verification_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

A11YD-04

A11YD-04 — Color Contrast & Visual Accessibility Spec

Header Block

- template_id: A11YD-04
- title: Color Contrast & Visual Accessibility Spec
- type: accessibility_design
- template_version: 1.0.0
- output_path: 10_app/accessibility/A11YD-04_Color_Contrast_Visual_A11y.md
- compliance_gate_id: TMP-05.PRIMARY.A11Y
- upstream_dependencies: ["DSYS-01"]
- inputs_required: ["DSYS-01", "DSYS-05", "RLB-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the visual accessibility requirements: contrast targets, acceptable combinations, focus visibility, and non-color cues. This ensures token systems and UI states remain accessible across themes and breakpoints.

Inputs Required

- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- DSYS-05: {{xref:DSYS-05}} | OPTIONAL
- RLB-01: {{xref:RLB-01}} | OPTIONAL

- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Contrast targets for:
    - normal text
    - large text
    - icons
    - interactive controls
    - focus indicators
- Approved color pairing rules (semantic token combinations)
- Non-color cue requirements (errors/success/warnings must have more than color)
- Focus visibility requirements (outline width, offset, visibility on all themes)
- Disabled state visibility rules (must still be perceivable)
- Verification method (tools + process)

Optional Fields

- High-contrast mode notes | OPTIONAL
- Brand exceptions (should be none for text) | OPTIONAL
- Notes | OPTIONAL

Rules

- Contrast requirements apply across all themes (DSYS-05).

- Never rely on color alone to convey meaning.
- Focus indicators must be visible on all backgrounds.
- Disabled controls must remain distinguishable from enabled.

Output Format

1) Contrast Targets (required)

element_type	target_ratio	notes
normal_text	{{targets.normal_text}}	{{targets.normal_notes}}
large_text	{{targets.large_text}}	{{targets.large_notes}}
icons	{{targets.icons}}	{{targets.icons_notes}}
controls	{{targets.controls}}	{{targets.controls_notes}}
focus_indicator	{{targets.focus}}	{{targets.focus_notes}}

2) Approved Pairings (required)

semantic_use	foreground_token	background_token	meets_target	notes
body_text_on_surface	{{pairs[0].fg}}	{{pairs[0].bg}}	{{pairs[0].meets}}	{{pairs[0].notes}}
primary_button_text	{{pairs[1].fg}}	{{pairs[1].bg}}	{{pairs[1].meets}}	{{pairs[1].notes}}

3) Non-Color Cues (required)

- Error indicators must include: {{noncolor.error}}
- Success indicators must include: {{noncolor.success}}
- Warning indicators must include: {{noncolor.warning}}

4) Focus Visibility (required)

- Focus outline rule: {{focus.outline_rule}}
- Offset rule: {{focus.offset_rule}} | OPTIONAL
- Focus on dark backgrounds: {{focus.dark_backgrounds}}
- Focus on light backgrounds: {{focus.light_backgrounds}}

5) Disabled State Rules (required)

- Disabled text: {{disabled.text}}
- Disabled controls: {{disabled.controls}}
- Disabled vs placeholder: {{disabled.placeholder_distinction}} | OPTIONAL

6) Verification Method (required)

- Tools: {{verify.tools}}
- Process: {{verify.process}}
- When required (PR reviews/releases): {{verify.when}} | OPTIONAL

Cross-References

- Upstream: {{xref:DSYS-01}} | OPTIONAL, {{xref:DSYS-05}} | OPTIONAL
- Downstream: {{xref:FE-06}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Targets + non-color cues + focus visibility basics.
- intermediate: Required. Approved pairings and disabled state rules.

- advanced: Required. Verification method and multi-theme coverage.

Unknown Handling

- UNKNOWN_ALLOWED: `high_contrast_notes`, `brand_exceptions` (should be none), `notes`, `pairings_notes`
- If contrast targets are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.A11Y
- Pass conditions:
  - required_fields_present == true
  - contrast_targets_present == true
  - approved_pairings_present == true
  - noncolor_cues_present == true
  - focus_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

A11YD-05

A11YD-05 — Accessible Error Messaging Rules

Header Block

- template_id: A11YD-05
- title: Accessible Error Messaging Rules
- type: accessibility_design
- template_version: 1.0.0
- output_path: 10_app/accessibility/A11YD-05_Accessible_Error_Messaging.md
- compliance_gate_id: TMP-05.PRIMARY.A11Y
- upstream_dependencies: ["DES-06", "DES-07", "CDX-04", "A11YD-02", "A11YD-03"]
- inputs_required: ["DES-06", "DES-07", "CDX-04", "A11YD-02", "A11YD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define enforceable rules for accessible error communication across forms and system errors: how errors are presented, announced, linked to fields, and recoverable. This ensures errors are understandable and operable for keyboard and screen reader users.

Inputs Required

- DES-06: {{xref:DES-06}} | OPTIONAL
- DES-07: {{xref:DES-07}} | OPTIONAL
- CDX-04: {{xref:CDX-04}} | OPTIONAL
- A11YD-02: {{xref:A11YD-02}} | OPTIONAL
- A11YD-03: {{xref:A11YD-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Error types covered:
  - field validation errors
  - form submission errors
  - permission/access errors
  - network/server errors

    - unknown errors
- Presentation rules per error type (inline/summary/banner/modal)
- Announcement rules per error type (when and how)
- Field association rules:
    - programmatic linking from error to field
    - “jump to error” behavior (optional but recommended)
- Focus rules:
    - focus on first error OR focus on error summary (deterministic)
    - focus restoration after correction
- Recovery guidance requirements (what user can do)
- Copy rules (structure + tone pointers)
- Verification checklist

Optional Fields

- Multi-error aggregation rules | OPTIONAL
- Localization notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Every error must answer: what happened, what it affects, what to do next.
- Errors must not rely on color alone; must include text and semantic markup.
- Inline errors must be announced when they appear or when focus enters the field (deterministic rule).
- If error summary exists, it must be keyboard navigable and link to each error field.
- Permission errors must not leak restricted info.

Output Format

1) Error Presentation Rules (required)

error_type	preferred_surface	when_used	must_include	user_actions
field_validation	{{rules.field.surface}}	{{rules.field.when}}	{{rules.field.must_include}}	{{rules.field.actions}}
form_submission	{{rules.form.surface}}	{{rules.form.when}}	{{rules.form.must_include}}	{{rules.form.actions}}
permission	{{rules.permission.surface}}	{{rules.permission.when}}	{{rules.permission.must_include}}	{{rules.permission.actions}}

network_server	{{rules.net.surface}}	{{rules.net.when}}	{{rules.net.must_include}}	{{rules.net.actions}}
unknown	{{rules.unknown.surface}}	{{rules.unknown.when}}	{{rules.unknown.must_include}}	{{rules.unknown.actions}}

2) Announcement Rules (required)

error_type	announce_when	mechanism	content_rule	notes
field_validation	{{announce.field.when}}	{{announce.field.mechanism}}	{{announce.field.content}}	{{announce.field.notes}}
form_submission	{{announce.form.when}}	{{announce.form.mechanism}}	{{announce.form.content}}	{{announce.form.notes}}
permission	{{announce.permission.when}}	{{announce.permission.mechanism}}	{{announce.permission.content}}	{{announce.permission.notes}}

3) Field Association Rules (required)

- Link error text to field: {{association.link_rule}}
- Error state markup rules: {{association.state_markup}} | OPTIONAL
- Jump-to-error behavior: {{association.jump_to_error}} | OPTIONAL

4) Focus Rules (required)

- Default focus behavior on submit: {{focus.on_submit}} (first_error / error_summary)
- Focus restoration after close (modal/banner): {{focus.restore}}
- Focus after correction: {{focus.after_fix}} | OPTIONAL

5) Recovery Guidance Rules (required)

- Required guidance structure: {{recovery.structure}}
- When to provide retry: {{recovery.retry_when}}
- When to provide support link: {{recovery.support_when}} | OPTIONAL

6) Copy Rules (required)

- Message structure: {{copy.structure}} (problem / impact / next step)
- Tone: {{xref:CDX-01}} | OPTIONAL
- Canonical messages: {{xref:CDX-04}} | OPTIONAL

7) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}}
- {{verify[4]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DES-06}} | OPTIONAL, {{xref:DES-07}} | OPTIONAL, {{xref:CDX-04}} | OPTIONAL
- Downstream: {{xref:FE-07}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define surfaces + focus rule + copy structure.
- intermediate: Required. Add announcements and field association rules.
- advanced: Required. Add jump-to-error behavior and multi-error aggregation policy.

Unknown Handling

- UNKNOWN_ALLOWED: `jump_to_error`, `multi_error_aggregation`, `localization_notes`, `notes`
- If focus.on_submit is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.A11Y
- Pass conditions:
  - required_fields_present == true
  - presentation_rules_present == true
  - announcement_rules_present == true
  - focus_rules_present == true
  - verification_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Responsive Layout & Breakpoints (RLB)

Responsive Layout & Breakpoints (RLB)

- RLB-01 Breakpoint Definitions (sizes + names)
- RLB-02 Layout Adaptation Rules (per breakpoint)
- RLB-03 Responsive Component Behavior (tables/cards/nav)
- RLB-04 Touch Target & Density Rules
- RLB-05 Responsive Media Rules (images/video scaling)

RLB-01

RLB-01 — Breakpoint Definitions (sizes + names)

Header Block

- template_id: RLB-01
- title: Breakpoint Definitions (sizes + names)
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-01_Breakpoint_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["DSYS-03", "DSYS-01"]
- inputs_required: ["DSYS-01", "DSYS-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical breakpoint system (names + pixel ranges) used across web and responsive surfaces. This document ensures consistent responsive behavior and prevents ad-hoc breakpoint usage.

Inputs Required

- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- DSYS-03: {{xref:DSYS-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Breakpoint list (minimum 4 for responsive web; justify if fewer)
- For each breakpoint:
    - bp_id
    - name
    - min_width_px
    - max_width_px (or open-ended)
    - intended devices/use cases
- Naming convention and stability rules
- Orientation notes (if applicable)
- Density scaling policy (if applicable)

Optional Fields

- Container query notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Breakpoint names must be stable; do not change semantics mid-project.
- Breakpoints must not overlap (except boundary edges).
- If mobile-native only, breakpoints may be N/A; mark explicitly.

Output Format

1) Breakpoints (canonical)

bp_id	name	min_width_px	max_width_px	intended_use	notes
bp_xs	{{bps.xs.name}}	{{bps.xs.min}}	{{bps.xs.max}}	{{bps.xs.use}}	{{bps.xs.notes}}
bp_sm	{{bps.sm.name}}	{{bps.sm.min}}	{{bps.sm.max}}	{{bps.sm.use}}	{{bps.sm.notes}}
bp_md	{{bps.md.name}}	{{bps.md.min}}	{{bps.md.max}}	{{bps.md.use}}	{{bps.md.notes}}
bp_lg	{{bps.lg.name}}	{{bps.lg.min}}	{{bps.lg.max}}	{{bps.lg.use}}	{{bps.lg.notes}}

2) Naming & Stability Rules (required)

- Naming convention: {{rules.naming}}
- Change policy: {{rules.change_policy}}

3) Orientation / Density Policy (optional)

- Orientation handling: {{rules.orientation}} | OPTIONAL
- Density scaling: {{rules.density}} | OPTIONAL

Cross-References

- Upstream: {{xref:DSYS-03}} | OPTIONAL
- Downstream: {{xref:RLB-02}}, {{xref:DSYS-05}} | OPTIONAL, {{xref:FE-01}} | OPTIONAL, {{xref:VAP-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define at least 4 breakpoints with clear ranges.
- intermediate: Required. Add intended use and stability rules.

- advanced: Required. Add orientation/density policy where relevant.

Unknown Handling

- UNKNOWN_ALLOWED: `orientation_policy`, `density_policy`, `container_query_notes`, `notes`
- If breakpoint ranges overlap → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE
- Pass conditions:
  - required_fields_present == true
  - breakpoints_count >= 4 (or justified)
  - breakpoint_ranges_non_overlapping == true
  - naming_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RLB-02

RLB-02 — Layout Adaptation Rules (per breakpoint)

Header Block

- template_id: RLB-02
- title: Layout Adaptation Rules (per breakpoint)
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-02_Layout_Adaptation_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["RLB-01", "DSYS-03", "DES-03"]
- inputs_required: ["RLB-01", "DSYS-03", "DES-03", "IAN-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic layout adaptation rules across breakpoints: how navigation, grids, density, and content hierarchy change as screen size changes. This ensures responsive behavior is predictable and implementation-ready.

Inputs Required

- RLB-01: {{xref:RLB-01}}
- DSYS-03: {{xref:DSYS-03}} | OPTIONAL
- DES-03: {{xref:DES-03}} | OPTIONAL

- IAN-01: {{xref:IAN-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Layout adaptation rules per breakpoint (at minimum for xs/sm/md/lg)
- For each breakpoint:
    - navigation adaptation (tabs/drawer/sidebar)
    - grid/columns adaptation
    - density adaptation (spacing adjustments)
    - content priority rules (what collapses, what stays visible)
    - component substitution rules (e.g., table → cards)
    - modal sizing rules (full-screen vs centered)
- Cross-breakpoint invariants (what must not change)
- Verification checklist

Optional Fields

- Platform-specific notes (web vs tablet) | OPTIONAL
- Notes | OPTIONAL

Rules

- Responsive changes must not break flow completion (DES-01).
- Information hierarchy must remain consistent; avoid hiding critical actions.

- Navigation changes must align to IAN-01 structure.
- Component substitution must preserve accessibility and semantics.

Output Format

1) Breakpoint Rules (canonical)

breakpoint	nav_pattern	grid_columns	density_mode	content_priority_rules	component_substitutions	modal_rules	notes
bp_xs	{{rules.xs.nav}}	{{rules.xs.cols}}	{{rules.xs.density}}	{{rules.xs.priority}}	{{rules.xs.subs}}	{{rules.xs.modal}}	{{rules.xs.notes}}
bp_sm	{{rules.sm.nav}}	{{rules.sm.cols}}	{{rules.sm.density}}	{{rules.sm.priority}}	{{rules.sm.subs}}	{{rules.sm.modal}}	{{rules.sm.notes}}
bp_md	{{rules.md.nav}}	{{rules.md.cols}}	{{rules.md.density}}	{{rules.md.priority}}	{{rules.md.subs}}	{{rules.md.modal}}	{{rules.md.notes}}
bp_lg	{{rules.lg.nav}}	{{rules.lg.cols}}	{{rules.lg.density}}	{{rules.lg.priority}}	{{rules.lg.subs}}	{{rules.lg.modal}}	{{rules.lg.notes}}

2) Cross-Breakpoint Invariants (required)

- {{invariants[0]}}
- {{invariants[1]}} | OPTIONAL
- {{invariants[2]}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}

- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RLB-01}}, {{xref:DSYS-03}} | OPTIONAL, {{xref:IAN-01}} | OPTIONAL
- Downstream: {{xref:RLB-03}}, {{xref:FE-01}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define nav and grid changes per breakpoint.
- intermediate: Required. Add substitutions and modal rules.
- advanced: Required. Add invariants and verification checklist.

Unknown Handling

- UNKNOWN_ALLOWED: `platform_notes`, `notes`, `density_mode` (if single mode)
- If any breakpoint lacks nav_pattern or content_priority_rules → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE
- Pass conditions:
    - required_fields_present == true
    - breakpoint_rules_present == true
    - invariants_present == true

- verification_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

RLB-03

RLB-03 — Responsive Component Behavior (tables/cards/nav)

Header Block

- template_id: RLB-03
- title: Responsive Component Behavior (tables/cards/nav)
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-03_Responsive_Component_Behavior.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["RLB-01", "DSYS-02", "DSYS-03", "A11YD-01"]
- inputs_required: ["RLB-01", "DSYS-02", "DSYS-03", "A11YD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how key UI components adapt across breakpoints and input types so responsive behavior is consistent and accessible. This covers component-level transformations (table → cards, nav shifts), density shifts, and interaction differences.

Inputs Required

- RLB-01: {{xref:RLB-01}} | OPTIONAL
- DSYS-02: {{xref:DSYS-02}} | OPTIONAL
- DSYS-03: {{xref:DSYS-03}} | OPTIONAL

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Component adaptation catalog (minimum 10 components/patterns)
- For each entry:
    - target_component_id or pattern name
    - breakpoint behavior (xs/sm/md/lg)
    - layout changes (stacking, wrapping, truncation)
    - interaction changes (hover vs tap, disclosure patterns)
    - accessibility impacts (focus order, SR semantics)
    - performance considerations (virtualization, image sizing)
- Navigation component behavior (tab/drawer/sidebar adaptations) if applicable
- Verification checklist

Optional Fields

- Platform notes (mobile web vs native) | OPTIONAL
- Notes | OPTIONAL

Rules

- Responsive adaptations must preserve meaning and available actions.
- If converting tables to cards, sorting/filtering must remain accessible.

- Truncation must not hide critical info; provide disclosure or expand.
- Keyboard navigation must still work in all breakpoint variants.

Output Format

1) Responsive Component Catalog (canonical)

target	type (component/pattern)	bp_xs	bp_sm	bp_md	bp_lg	interaction_changes	a11y_notes	perf_notes	notes
{{items[0].target}}	{{items[0].type}}	{{items[0].xs}}	{{items[0].sm}}	{{items[0].md}}	{{items[0].lg}}	{{items[0].interaction}}	{{items[0].a11y}}	{{items[0].perf}}	{{items[0].notes}}
{{items[1].target}}	{{items[1].type}}	{{items[1].xs}}	{{items[1].sm}}	{{items[1].md}}	{{items[1].lg}}	{{items[1].interaction}}	{{items[1].a11y}}	{{items[1].perf}}	{{items[1].notes}}

2) Navigation Component Behavior (required if responsive web)

- Tabs → drawer rules: {{nav.tabs_to_drawer}} | OPTIONAL
- Sidebar collapse rules: {{nav.sidebar_collapse}} | OPTIONAL
- Breadcrumbs behavior: {{nav.breadcrumbs}} | OPTIONAL

3) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RLB-01}} | OPTIONAL, {{xref:DSYS-02}} | OPTIONAL, {{xref:DSYS-03}} | OPTIONAL
- Downstream: {{xref:RLB-04}}, {{xref:FE-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define 10 responsive component behaviors.
- intermediate: Required. Add interaction differences and a11y notes.
- advanced: Required. Add perf notes and verification checklist.

Unknown Handling

- UNKNOWN_ALLOWED: `platform_notes`, `nav_component_behavior` (if not applicable), `notes`
- If any component lacks breakpoint behavior definition → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE
- Pass conditions:
    - required_fields_present == true
    - items_count >= 10
    - breakpoint_behavior_complete == true
    - verification_present == true
    - placeholder_resolution == true

    - no_unapproved_unknowns == true

RLB-04 — Touch Target & Density Rules

Header Block

- template_id: RLB-04
- title: Touch Target & Density Rules
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-04_Touch_Target_Density_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["A11YD-01", "DSYS-03", "RLB-01"]
- inputs_required: ["A11YD-01", "DSYS-03", "RLB-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the rules for touch target sizing, spacing, and density modes across devices so UI remains usable on touch and pointer inputs. This prevents overly dense layouts that break usability and accessibility.

Inputs Required

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- DSYS-03: {{xref:DSYS-03}} | OPTIONAL

- RLB-01: {{xref:RLB-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Minimum touch target size rule (width/height)
- Minimum spacing between interactive targets
- Density modes:
    - comfortable
    - compact (optional)
- Rules for when compact mode is allowed
- Pointer vs touch differences (hover availability, hit slop)
- Verification checklist

Optional Fields

- Per-component exceptions | OPTIONAL
- Notes | OPTIONAL

Rules

- Touch target minimums apply regardless of theme.
- If compact mode is enabled, it must not reduce touch targets below minimum.
- Provide hit slop guidance on mobile where targets are visually small.
- Must align with DSYS spacing rules.

Output Format

1) Minimum Target Rules (required)

- Minimum target size: {{targets.min_size}}
- Minimum spacing: {{targets.min_spacing}}
- Hit slop guidance (mobile): {{targets.hit_slop}} | OPTIONAL

2) Density Modes (required)

mode	description	allowed_surfaces	constraints
comfortable	{{density.comfortable.desc}}	{{density.comfortable.surfaces}}	{{density.comfortable.constraints}}
compact	{{density.compact.desc}}	{{density.compact.surfaces}}	{{density.compact.constraints}}

3) Pointer vs Touch Rules (required)

- Hover availability assumption: {{input.hover_assumption}}
- Tap vs click equivalence: {{input.tap_click_equivalence}}
- Long-press rules (if used): {{input.long_press}} | OPTIONAL

4) Exceptions (optional)

exception_id	component_or_screen	exception	rationale	mitigation
ex_01	{{exceptions[0].target}}	{{exceptions[0].exception}}	{{exceptions[0].rationale}}	{{exceptions[0].mitigation}}

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}

- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:A11YD-01}} | OPTIONAL, {{xref:DSYS-03}} | OPTIONAL, {{xref:RLB-01}} | OPTIONAL
- Downstream: {{xref:RLB-05}}, {{xref:FE-*}}* | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Min target + spacing + density definition.
- intermediate: Required. Add pointer vs touch rules and verification.
- advanced: Required. Add exception governance and hit slop guidance.

Unknown Handling

- UNKNOWN_ALLOWED: `compact_mode`, `exceptions`, `long_press_rules`, `notes`
- If minimum target size is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE
- Pass conditions:
    - required_fields_present == true

- min_target_rules_present == true
- density_modes_present == true
- verification_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

RLB-04

RLB-04 — Touch Target & Density Rules

Header Block

- template_id: RLB-04
- title: Touch Target & Density Rules
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-04_Touch_Target_Density_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["A11YD-01", "DSYS-03", "RLB-01"]
- inputs_required: ["A11YD-01", "DSYS-03", "RLB-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the rules for touch target sizing, spacing, and density modes across devices so UI remains usable on touch and pointer inputs. This prevents overly dense layouts that break usability and accessibility.

Inputs Required

- A11YD-01: {{xref:A11YD-01}} | OPTIONAL
- DSYS-03: {{xref:DSYS-03}} | OPTIONAL
- RLB-01: {{xref:RLB-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Minimum touch target size rule (width/height)
- Minimum spacing between interactive targets
- Density modes:
    - comfortable
    - compact (optional)
- Rules for when compact mode is allowed
- Pointer vs touch differences (hover availability, hit slop)
- Verification checklist

Optional Fields

- Per-component exceptions | OPTIONAL
- Notes | OPTIONAL

Rules

- Touch target minimums apply regardless of theme.
- If compact mode is enabled, it must not reduce touch targets below minimum.
- Provide hit slop guidance on mobile where targets are visually small.
- Must align with DSYS spacing rules.

Output Format

1) Minimum Target Rules (required)

- Minimum target size: {{targets.min_size}}
- Minimum spacing: {{targets.min_spacing}}
- Hit slop guidance (mobile): {{targets.hit_slop}} | OPTIONAL

2) Density Modes (required)

mode	description	allowed_surfaces	constraints
comfortable	{{density.comfortable.desc}}	{{density.comfortable.surfaces}}	{{density.comfortable.constraints}}
compact	{{density.compact.desc}}	{{density.compact.surfaces}}	{{density.compact.constraints}}

3) Pointer vs Touch Rules (required)

- Hover availability assumption: {{input.hover_assumption}}
- Tap vs click equivalence: {{input.tap_click_equivalence}}
- Long-press rules (if used): {{input.long_press}} | OPTIONAL

4) Exceptions (optional)

exception_id	component_or_screen	exception	rationale	mitigation
ex_01	{{exceptions[0].target}}	{{exceptions[0].exception}}	{{exceptions[0].rationale}}	{{exceptions[0].mitigation}}

5) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:A11YD-01}} | OPTIONAL, {{xref:DSYS-03}} | OPTIONAL, {{xref:RLB-01}} | OPTIONAL
- Downstream: {{xref:RLB-05}}, {{xref:FE-*}} | *OPTIONAL, {{xref:MOB-}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Min target + spacing + density definition.
- intermediate: Required. Add pointer vs touch rules and verification.
- advanced: Required. Add exception governance and hit slop guidance.

Unknown Handling

- UNKNOWN_ALLOWED: `compact_mode`, `exceptions`, `long_press_rules`, `notes`
- If minimum target size is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE
- Pass conditions:
    - required_fields_present == true
    - min_target_rules_present == true
    - density_modes_present == true
    - verification_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

RLB-05

RLB-05 — Responsive Media Rules (images/video scaling)

Header Block

- template_id: RLB-05
- title: Responsive Media Rules (images/video scaling)
- type: responsive_layout_breakpoints
- template_version: 1.0.0
- output_path: 10_app/responsive/RLB-05_Responsive_Media_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.RESPONSIVE
- upstream_dependencies: ["RLB-01", "VAP-02", "A11YD-01"]
- inputs_required: ["RLB-01", "VAP-02", "A11YD-01", "DSYS-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define how media (images, video, icons, thumbnails) scales and behaves across breakpoints and device densities. This ensures consistent quality, performance, and accessibility for responsive UI.

Inputs Required

- RLB-01: {{xref:RLB-01}} | OPTIONAL
- VAP-02: {{xref:VAP-02}} | OPTIONAL
- A11YD-01: {{xref:A11YD-01}} | OPTIONAL

- DSYS-03: {{xref:DSYS-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Media types covered (images, avatars, thumbnails, video, icons, illustrations)
- Scaling rules:
    - aspect ratio handling (crop vs letterbox)
    - container-fit rules (cover/contain equivalents)
    - max/min sizes by breakpoint class
- Device density rules (1x/2x/3x) and selection logic
- Loading rules:
    - placeholders/skeletons
    - progressive loading (if used)
    - error fallbacks
- Performance rules:
    - compression targets guidance
    - lazy-load rules
    - avoid layout shift rules
- Accessibility rules:
    - alt text requirements
    - captions/subtitles for video (if applicable)
- Verification checklist

Optional Fields

- Content moderation/blur rules (if applicable) | OPTIONAL
- Notes | OPTIONAL

Rules

- Media must not cause layout shift; define reserved space behavior.
- Cropping rules must not hide critical information without fallback.
- Video must have accessible controls and captions if content requires.
- Image density selection must be deterministic and match asset export spec (VAP-02).

Output Format

1) Media Scaling Rules (required)

media_type	aspect_ratio_policy	fit_policy	max_size_by_bp	min_size_by_bp	notes
image	{{rules.image.aspect}}	{{rules.image.fit}}	{{rules.image.max}}	{{rules.image.min}}	{{rules.image.notes}}
thumbnail	{{rules.thumb.aspect}}	{{rules.thumb.fit}}	{{rules.thumb.max}}	{{rules.thumb.min}}	{{rules.thumb.notes}}
video	{{rules.video.aspect}}	{{rules.video.fit}}	{{rules.video.max}}	{{rules.video.min}}	{{rules.video.notes}}

2) Density Selection (required)

- Supported densities: {{density.supported}} (1x/2x/3x)
- Selection logic: {{density.selection_logic}}
- Asset source: {{xref:VAP-02}} | OPTIONAL

3) Loading + Error Rules (required)

- Placeholder policy: {{loading.placeholder}}
- Progressive loading: {{loading.progressive}} | OPTIONAL
- Error fallback behavior: {{loading.error_fallback}}
- Retry behavior: {{loading.retry}} | OPTIONAL

4) Performance Rules (required)

- Compression guidance: {{perf.compression}}
- Lazy-load rule: {{perf.lazy_load}}
- Layout shift prevention: {{perf.no_layout_shift}}

5) Accessibility Rules (required)

- Alt text rule: {{a11y.alt_text}}
- Decorative images: {{a11y.decorative}}
- Video captions/subtitles: {{a11y.captions}} | OPTIONAL
- Media controls a11y: {{a11y.controls}} | OPTIONAL

6) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:RLB-01}} | OPTIONAL, {{xref:VAP-02}} | OPTIONAL, {{xref:A11YD-01}} | OPTIONAL

- Downstream: {{xref:FE-*}}* | *OPTIONAL, {{xref:FPMP-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL

- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Scaling rules + density selection basics + alt text rule.

- intermediate: Required. Add loading/error and performance rules.

- advanced: Required. Add video accessibility and verification checklist rigor.

Unknown Handling

- UNKNOWN_ALLOWED: `progressive_loading`, `captions`, `controls_a11y`, `moderation_rules`, `notes`

- If density selection logic is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESPONSIVE

- Pass conditions:

    - required_fields_present == true

    - scaling_rules_present == true

    - density_rules_present == true

    - loading_error_rules_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true

Visual Asset Production (VAP)

Visual Asset Production (VAP)

- VAP-01 Asset Inventory (logos, icons, illustrations)
- VAP-02 Export Spec (formats, sizes, naming, density)
- VAP-03 Brand Usage Rules (do/don’t, clearspace)
- VAP-04 Asset Delivery Checklist (handoff requirements)
- VAP-05 Accessibility for Visual Assets (alt text, meaning)

VAP-01

VAP-01 — Asset Inventory (logos, icons, illustrations)

Header Block

- template_id: VAP-01
- title: Asset Inventory (logos, icons, illustrations)
- type: visual_asset_production
- template_version: 1.0.0
- output_path: 10_app/assets/VAP-01_Asset_Inventory.md
- compliance_gate_id: TMP-05.PRIMARY.ASSETS
- upstream_dependencies: ["DSYS-04", "CDX-03", "DSYS-01"]
- inputs_required: ["DSYS-04", "CDX-03", "DSYS-01", "IAN-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical inventory of visual assets required for the product (logos, icons, illustrations, images, animations). This inventory drives deterministic production, export, naming, and handoff so implementation does not invent assets.

Inputs Required

- DSYS-04: {{xref:DSYS-04}} | OPTIONAL
- CDX-03: {{xref:CDX-03}} | OPTIONAL
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL

- IAN-01: {{xref:IAN-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Asset list (minimum 20 for non-trivial products; justify if smaller)
- For each asset:
    - asset_id
    - asset_type (logo/icon/illustration/photo/animation)
    - name
    - purpose/where used (screen_id/component_id)
    - required sizes (px) and densities (1x/2x/3x) if applicable
    - format (svg/png/webp/mp4/etc.)
    - theme variants needed (light/dark) (if applicable)
    - accessibility classification (decorative/informative)
    - alt text requirement (if informative)
    - ownership (who produces/approves)
    - status (needed/in_progress/done)
    - delivery path (output location) | OPTIONAL

Optional Fields

- Source file pointers (figma links, source repo) | OPTIONAL
- Licensing notes (for photos/icons) | OPTIONAL

- Notes | OPTIONAL

Rules

- Asset IDs must be stable and unique (`asset_<slug>`).
- If an asset is informative, alt text requirement must be specified.
- If multiple themes exist, required theme variants must be specified for relevant assets.
- Formats must align with DSYS-04 style rules and responsive rules (RLB-05/VAP-02).

Output Format

1) Asset Inventory (canonical)

asset_id	type	name	used_in	sizes	densities	format	theme_variants	a11y_class	alt_text_req	owner	status	delivery_path	notes
asset_001	{{assets[0].type}}	{{assets[0].name}}	{{assets[0].used_in}}	{{assets[0].sizes}}	{{assets[0].densities}}	{{assets[0].format}}	{{assets[0].themes}}	{{assets[0].a11y_class}}	{{assets[0].alt_req}}	{{assets[0].owner}}	{{assets[0].status}}	{{assets[0].path}}	{{assets[0].notes}}
asset_	{{assets[1].type}}	{{assets[1].name}}	{{assets[1].used_in}}	{{assets[1].sizes}}	{{assets[1].densities}}	{{assets[1].format}}	{{assets[1].themes}}	{{assets[1].a11y_class}}	{{assets[1].alt_req}}	{{assets[1].owner}}	{{assets[1].status}}	{{assets[1].path}}	{{assets[1].notes}}

0
0
2

2) Coverage Checks (required)

- Assets exist for all referenced empty states/onboarding visuals: {{coverage.empty_onboarding}}
- Icon inventory aligns with DSYS-04 rules: {{coverage.icon_rules_alignment}}
- Theme variants listed where required: {{coverage.theme_variants_complete}}

Cross-References

- Upstream: {{xref:DSYS-04}} | OPTIONAL, {{xref:CDX-03}} | OPTIONAL
- Downstream: {{xref:VAP-02}}, {{xref:VAP-04}} | OPTIONAL, {{xref:RLB-05}} | OPTIONAL, {{xref:FE-*}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 20 assets with types + used_in + format.
- intermediate: Required. Add sizes/densities and a11y classification.
- advanced: Required. Add theme variants and coverage checks.

Unknown Handling

- UNKNOWN_ALLOWED: `delivery_path`, `source_pointers`, `licensing_notes`, `notes`, `densities` (if not applicable)
- If a11y_class is UNKNOWN for any asset → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ASSETS
- Pass conditions:
  - required_fields_present == true
  - assets_count >= 20 (or justified)
  - a11y_class_complete == true
  - coverage_checks_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

VAP-02

VAP-02 — Export Spec (formats, sizes, naming, density)

Header Block

- template_id: VAP-02
- title: Export Spec (formats, sizes, naming, density)
- type: visual_asset_production
- template_version: 1.0.0
- output_path: 10_app/assets/VAP-02_Export_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.ASSETS
- upstream_dependencies: ["VAP-01", "DSYS-04", "RLB-05"]
- inputs_required: ["VAP-01", "DSYS-04", "RLB-05", "DSYS-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define deterministic export and delivery rules for visual assets: formats, sizes, densities, naming conventions, and folder layout. This ensures implementation can consume assets without manual guessing or inconsistent exports.

Inputs Required

- VAP-01: {{xref:VAP-01}}
- DSYS-04: {{xref:DSYS-04}} | OPTIONAL
- RLB-05: {{xref:RLB-05}} | OPTIONAL

- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Naming convention rules (file names, casing, separators)
- Folder organization rules (by type, by feature/screen, by theme)
- Export formats by asset type (svg/png/webp/jpg/mp4/etc.)
- Size rules:
    - base size(s)
    - responsive variants
    - max size constraints
- Density rules:
    - which densities required (1x/2x/3x)
    - when to omit densities (svg)
- Theme variant rules (light/dark)
- Optimization rules (compression targets, stripping metadata)
- Verification checklist (how to verify exports match spec)

Optional Fields

- CDN/hosting notes | OPTIONAL
- Source-of-truth tool notes (Figma) | OPTIONAL
- Notes | OPTIONAL

Rules

- Export rules must match responsive media rules (RLB-05).
- SVG assets must be optimized and sanitized (no embedded raster unless allowed).
- Raster assets must meet compression targets.
- Filenames must be deterministic and consistent with asset_id mapping.

Output Format

1) Naming Rules (required)

- File naming convention: {{naming.convention}} (e.g., <asset_id>**<variant>**<size>@<density>.<ext>)
- Casing: {{naming.casing}}
- Separator rules: {{naming.separators}}
- Theme suffix rules: {{naming.theme_suffix}} | OPTIONAL

2) Folder Organization (required)

- Base folder: {{folders.base}}
- By type: {{folders.by_type}}
- By theme: {{folders.by_theme}} | OPTIONAL
- By feature/screen: {{folders.by_feature}} | OPTIONAL

3) Export Formats (required)

asset_type	preferred_formats	allowed_formats	notes
icon	{{formats.icon.preferred}}	{{formats.icon.allowed}}	{{formats.icon.notes}}

illustration	{{formats.illustration.preferred}}	{{formats.illustration.allowed}}	{{formats.illustration.notes}}
photo	{{formats.photo.preferred}}	{{formats.photo.allowed}}	{{formats.photo.notes}}
animation	{{formats.animation.preferred}}	{{formats.animation.allowed}}	{{formats.animation.notes}}

4) Sizes & Densities (required)

- Base sizes policy: {{sizes.base_policy}}
- Responsive variants policy: {{sizes.responsive_policy}} | OPTIONAL
- Densities required: {{densities.required}}
- When to omit densities: {{densities.omit_when}}

5) Theme Variants (required)

- When light/dark exports required: {{themes.when_required}}
- Naming mapping for theme: {{themes.naming_mapping}} | OPTIONAL

6) Optimization Rules (required)

- Compression targets: {{opt.compression_targets}}
- Metadata stripping: {{opt.strip_metadata}}
- SVG sanitization: {{opt.svg_sanitize}} | OPTIONAL

7) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}

- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:VAP-01}}, {{xref:RLB-05}} | OPTIONAL, {{xref:DSYS-04}} | OPTIONAL
- Downstream: {{xref:VAP-04}}, {{xref:FE-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Naming + formats + densities + folder layout.
- intermediate: Required. Add optimization rules and theme variants.
- advanced: Required. Add verification checklist and responsive variant guidance.

Unknown Handling

- UNKNOWN_ALLOWED: `cdn_notes`, `source_tool_notes`, `responsive_variants`, `notes`
- If naming convention is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ASSETS
- Pass conditions:
    - required_fields_present == true
    - naming_rules_present == true
    - formats_present == true

- densities_rules_present == true
- optimization_rules_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

VAP-03

VAP-03 — Brand Usage Rules (do/don’t, clearspace)

Header Block

- template_id: VAP-03
- title: Brand Usage Rules (do/don’t, clearspace)
- type: visual_asset_production
- template_version: 1.0.0
- output_path: 10_app/assets/VAP-03_Brand_Usage_Rules.md
- compliance_gate_id: TMP-05.PRIMARY.ASSETS
- upstream_dependencies: ["VAP-01", "DSYS-01", "DSYS-05"]
- inputs_required: ["VAP-01", "DSYS-01", "DSYS-05", "CDX-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the enforceable rules for brand asset usage across the product so logos and brand visuals remain consistent, legible, and compliant. This includes clearspace, minimum sizes, acceptable backgrounds, and do/don’t examples.

Inputs Required

- VAP-01: {{xref:VAP-01}} | OPTIONAL
- DSYS-01: {{xref:DSYS-01}} | OPTIONAL
- DSYS-05: {{xref:DSYS-05}} | OPTIONAL

- CDX-01: {{xref:CDX-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Logo variants list (primary/mono/icon-only/etc.)
- For each logo variant:
    - variant_id
    - file reference (asset_id)
    - minimum size rules
    - clearspace rules
    - acceptable background rules
    - forbidden usage rules
- Brand color usage rules (ties to DSYS tokens)
- Typography usage rules (ties to DSYS tokens)
- Do/Don’t examples (minimum 10)
- Approval workflow (who approves brand exceptions)

Optional Fields

- Co-branding rules | OPTIONAL
- Legal trademark notes | OPTIONAL
- Notes | OPTIONAL

Rules

- Brand asset usage must not break accessibility (contrast, legibility).
- Logo must not be stretched, skewed, recolored outside allowed variants.
- Clearspace must be maintained in all placements.
- Any exception requires approval and must be logged (STK).

Output Format

1) Logo Variants (required)

variant_id	asset_id	min_size	clearspace_rule	allowed_backgrounds	forbidden_use	notes
logo_primary	{{logos.primary.asset_id}}	{{logos.primary.min_size}}	{{logos.primary.clearspace}}	{{logos.primary.backgrounds}}	{{logos.primary.forbidden}}	{{logos.primary.notes}}
logo_mono	{{logos.mono.asset_id}}	{{logos.mono.min_size}}	{{logos.mono.clearspace}}	{{logos.mono.backgrounds}}	{{logos.mono.forbidden}}	{{logos.mono.notes}}

2) Brand Color Usage (required)

- Token source: {{xref:DSYS-01}} | OPTIONAL
- Allowed primary usage: {{brand_colors.allowed_primary}}
- Forbidden color usage: {{brand_colors.forbidden}}
- Background rules: {{brand_colors.background_rules}} | OPTIONAL

3) Typography Usage (required)

- Token source: {{xref:DSYS-01}} | OPTIONAL
- Heading rules: {{typography.headings}}
- Body rules: {{typography.body}}

- Forbidden substitutions: {{typography.forbidden}} | OPTIONAL

4) Do/Don’t Examples (required, min 10)

example_id	do	dont	why
ex_01	{{examples[0].do}}	{{examples[0].dont}}	{{examples[0].why}}
ex_02	{{examples[1].do}}	{{examples[1].dont}}	{{examples[1].why}}

5) Approval Workflow (required)

- Default approver: {{approval.approver}}
- What requires approval: {{approval.requires}}
- Logging requirement: {{approval.logging}} | OPTIONAL

Cross-References

- Upstream: {{xref:VAP-01}} | OPTIONAL, {{xref:DSYS-01}} | OPTIONAL, {{xref:DSYS-05}} | OPTIONAL
- Downstream: {{xref:VAP-04}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Logo variants + min size + clearspace + do/don’t list.
- intermediate: Required. Add color/typography token usage rules.
- advanced: Required. Add approval workflow and exception logging rules.

Unknown Handling

- UNKNOWN_ALLOWED: `co_branding`, `legal_notes`, `notes`, `approval.logging`
- If clearspace rules are UNKNOWN for any logo variant → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ASSETS
- Pass conditions:
    - required_fields_present == true
    - logo_variants_present == true
    - do_dont_examples_count >= 10
    - approval_workflow_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

VAP-04

VAP-04 — Asset Delivery Checklist (handoff requirements)

Header Block

- template_id: VAP-04
- title: Asset Delivery Checklist (handoff requirements)
- type: visual_asset_production
- template_version: 1.0.0
- output_path: 10_app/assets/VAP-04_Asset_Delivery_Checklist.md
- compliance_gate_id: TMP-05.PRIMARY.ASSETS
- upstream_dependencies: ["VAP-01", "VAP-02"]
- inputs_required: ["VAP-01", "VAP-02", "RLB-05", "A11YD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the deterministic handoff checklist for delivering assets to engineering: what must be provided, where it goes, how it is verified, and what “done” means. This prevents incomplete or inconsistent asset deliveries.

Inputs Required

- VAP-01: {{xref:VAP-01}} | OPTIONAL
- VAP-02: {{xref:VAP-02}} | OPTIONAL
- RLB-05: {{xref:RLB-05}} | OPTIONAL

- A11YD-03: {{xref:A11YD-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Delivery package definition (folders + files expected)
- Checklist items (minimum 20)
- Verification steps (minimum 8)
- For each delivered asset set:
    - asset_ids included
    - naming compliance check
    - density coverage check
    - theme coverage check (if applicable)
    - accessibility coverage (alt text/classification)
    - optimization check (size/compression)
    - version tagging rule (if applicable)
- Acceptance criteria for “handoff complete”

Optional Fields

- Release milestone mapping | OPTIONAL
- Notes | OPTIONAL

Rules

- Handoff is not complete unless naming and export specs are met (VAP-02).
- Accessibility metadata must be included for informative assets (alt/labels).
- Engineering must have deterministic paths and keys for consumption.

Output Format

1) Delivery Package (required)

- Base path: {{delivery.base_path}}
- Folder structure: {{delivery.folder_structure}}
- Included file types: {{delivery.file_types}}

2) Checklist (required, min 20)

- {{checklist[0]}}
- {{checklist[1]}}
- {{checklist[2]}}
- {{checklist[3]}}
- {{checklist[4]}}
- {{checklist[5]}} | OPTIONAL

3) Verification Steps (required, min 8)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}}

- {{verify[4]}}
- {{verify[5]}} | OPTIONAL

4) Asset Set Validation Table (required)

asset_set_id	asset_ids	naming_ok	densities_ok	themes_ok	a11y_ok	optimization_ok	version_tag	notes
set_01	{{sets[0].asset_ids}}	{{sets[0].naming_ok}}	{{sets[0].densities_ok}}	{{sets[0].themes_ok}}	{{sets[0].a11y_ok}}	{{sets[0].opt_ok}}	{{sets[0].version_tag}}	{{sets[0].notes}}

5) Handoff Complete Criteria (required)

- All VAP-01 assets delivered or explicitly deferred: {{done.all_assets_accounted}}
- Naming/export spec compliance: {{done.spec_compliance}}
- A11y metadata complete: {{done.a11y_complete}}
- Verified by: {{done.verified_by}}

Cross-References

- Upstream: {{xref:VAP-01}} | OPTIONAL, {{xref:VAP-02}} | OPTIONAL, {{xref:RLB-05}} | OPTIONAL
- Downstream: {{xref:FE-*}}* | *OPTIONAL, {{xref:FPMP-*}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Checklist + delivery package + done criteria.

- intermediate: Required. Add validation table and verification steps.
- advanced: Required. Add version tagging and milestone mapping.

Unknown Handling

- UNKNOWN_ALLOWED: `milestone_mapping`, `version_tagging`, `notes`
- If delivery.base_path is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ASSETS
- Pass conditions:
    - required_fields_present == true
    - checklist_count >= 20
    - verification_steps_count >= 8
    - asset_set_validation_present == true
    - done_criteria_present == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

VAP-05

VAP-05 — Accessibility for Visual Assets (alt text, meaning)

Header Block

- template_id: VAP-05
- title: Accessibility for Visual Assets (alt text, meaning)
- type: visual_asset_production
- template_version: 1.0.0
- output_path: 10_app/assets/VAP-05_Accessibility_for_Visual_Assets.md
- compliance_gate_id: TMP-05.PRIMARY.ASSETS
- upstream_dependencies: ["A11YD-03", "DSYS-04", "VAP-01"]
- inputs_required: ["A11YD-03", "DSYS-04", "VAP-01", "CDX-01", "CDX-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define enforceable accessibility rules for visual assets: alt text, decorative vs informative classification, icon-only control labeling, and how meaning is conveyed. This ensures assets don’t create accessibility gaps or confuse assistive technologies.

Inputs Required

- A11YD-03: {{xref:A11YD-03}} | OPTIONAL
- DSYS-04: {{xref:DSYS-04}} | OPTIONAL
- VAP-01: {{xref:VAP-01}} | OPTIONAL

- CDX-01: {{xref:CDX-01}} | OPTIONAL
- CDX-02: {{xref:CDX-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Classification rules:
    - decorative assets
    - informative assets
    - functional assets (icons used as controls)
- Alt text requirements:
    - when required
    - length guidance
    - content rules (what to say / what not to say)
    - localization readiness
- Icon-only control rules:
    - accessible label source (CDX vs custom)
    - when tooltips are acceptable (not a substitute)
- Illustration meaning rules:
    - ensure copy conveys meaning, not illustration alone
    - avoid culturally specific signals unless localized
- Inventory compliance:
    - every VAP-01 asset has a11y_class

    - informative assets have alt_text requirement

- Verification checklist

Optional Fields

- Audio description rules (video) | OPTIONAL
- Notes | OPTIONAL

Rules

- Decorative assets must be hidden from screen readers.
- Informative assets must have alt text that conveys purpose, not appearance only.
- Icons used as controls must have programmatic labels even if visually unlabeled.
- Alt text must not duplicate adjacent visible text unless needed for context.

Output Format

1) Classification Rules (required)

class	definition	examples	SR_behavior
decorative	{{classes.decorative.def}}	{{classes.decorative.examples}}	{{classes.decorative.sr_behavior}}
informative	{{classes.informative.def}}	{{classes.informative.examples}}	{{classes.informative.sr_behavior}}
functional	{{classes.functional.def}}	{{classes.functional.examples}}	{{classes.functional.sr_behavior}}

2) Alt Text Rules (required)

- When required: {{alt.when_required}}

- Length guidance: {{alt.length}}
- Content rules (do): {{alt.do_rules}}
- Content rules (don't): {{alt.dont_rules}}
- Localization readiness: {{alt.l10n}}

3) Icon-only Controls (required)

- Label source preference: {{icon_controls.label_source}} (CDX-02 key vs explicit label)
- Tooltip policy: {{icon_controls.tooltip_policy}}
- Disallowed: {{icon_controls.disallowed}} | OPTIONAL

4) Illustration Meaning Rules (required)

- Illustration never sole meaning: {{illustrations.not_sole_meaning}}
- Empty state illustration pairing rule: {{illustrations.empty_state_pairing}} | OPTIONAL

5) Inventory Compliance Checks (required)

- All assets have a11y_class: {{checks.all_assets_classified}}
- Informative assets have alt requirements: {{checks.informative_have_alt}}
- Functional icons have labels: {{checks.functional_have_labels}}

6) Verification Checklist (required)

- {{verify[0]}}
- {{verify[1]}}
- {{verify[2]}}
- {{verify[3]}} | OPTIONAL

Cross-References

- Upstream: {{xref:VAP-01}} | OPTIONAL, {{xref:A11YD-03}} | OPTIONAL, {{xref:DSYS-04}} | OPTIONAL
- Downstream: {{xref:FE-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-A11Y]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Classification + alt rules + icon-label rules.
- intermediate: Required. Add inventory compliance checks and illustration rules.
- advanced: Required. Add verification checklist rigor and localization constraints.

Unknown Handling

- UNKNOWN_ALLOWED: `audio_description_rules`, `notes`, `disallowed`
- If classification rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ASSETS
- Pass conditions:
    - required_fields_present == true
    - classification_rules_present == true
    - alt_text_rules_present == true
    - inventory_checks_present == true

- placeholder_resolution == true
- no_unapproved_unknowns == true